



OPEN Integrating Lyapunov based backstepping and neuro fuzzy logic with sliding mode control for precise trajectory tracking of differential drive robots

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This paper presents the Fixed Ultra Hybrid Adaptive Controller (FUHAC) to the differential drive mobile robots to realize proper trajectory tracking in dynamic uncertainties, non-linearities, and external disturbances. FUHAC combines a number of nonlinear clustering methods, including adaptive backstepping, neural fuzzy inference, sliding mode compensation, disturbance observation and predictive error anticipation, into a multi-rate, adaptive gain scheduling model, which is Lyapunov stable. The tracking error, adaptive weighting dynamics and disturbance dynamics are coupled together to form a composite Lyapunov function, which guarantees Global Uniform Ultimate Boundedness (GUUB); when the approximation residuals are small enough, Global Asymptotic Stability (GAS) can be achieved. Three benchmark paths of lemniscate, circle, and diamond were simulated in large scale. The final error of position was less than 4 cm on all the trajectories. ISE/IAE/ITAE = (3.70/4.70/21.98) of the lemniscate, (1.08/2.62/18.81) of the circle, and (1.20/3.83/35.67) of the diamond was reported as the cumulative performance indices. Actuator torque mean was less than 10Nm, and settling time was 12.7s. Oscillation waveforms remained within 0.95 in all conditions, and the stabilized adaptive sliding mode is found to stabilize at $K_s = 3.7\text{--}4.6$, which confirms energy efficient stability maintenance. The controller was also tested on publicly available data on Pioneer 1 time series. Whereas the closed loop was also stable with a return to maneuverability in the aggressive maneuvers, practical errors in tracking were larger than those in simulation: average positional error 1.47 m, maximum instantaneous deviation 6.44 m, final position error 2.33 m, and RMS error 1.85 m. These transients were associated with actuator saturation around rated torque limits and extensive magnitude low frequency control actions. On the whole, FUHAC provides high performance, globally stable and computationally-efficient control of autonomous ground vehicles under uncertain or time varying conditions.

Keywords Mobile robot, Disturbance observer, Lyapunov-guaranteed, Backstepping, Neuro-fuzzy inference

The rapid advancement of autonomous mobile robots in industrial logistics, service sectors, and hazardous environments has ramped up the need for robust and highly precise trajectory-tracking methods^{1–4}. One of the most popular platforms for these uses is the differential-drive mobile robot (DDMR), which is appreciated for its simple mechanics and high maneuverability. Lately, it's been upgraded with features like adjustable wheelbases, bio-inspired soft movements, or adhesive locomotion^{5–8}. That said, DDMRs come with built-in nonholonomic constraints, plus issues like parametric uncertainties, unmodeled dynamics, and external disturbances, all of which make controller design super challenging^{9–11}. Traditional control techniques often fall short in broad operating conditions, which is why we're seeing a push toward more sophisticated control approaches that can adapt to changing environments while keeping the closed-loop system stable.

Recent progress in nonlinear control theory has brought some cool strategies for mobile robot navigation. A solid way to handle nonlinear systems is through backstepping, which gives a structured, step-by-step design

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with solid stability proofs^{10,12–15}. However, classic backstepping controllers tend to struggle when there are heavy uncertainties or disturbances. On the other hand, smart control methods using neural networks, fuzzy logic, reinforcement learning, and adaptive dynamic programming have shown they can approximate unknown nonlinearities and adjust online to shifting conditions^{11,16–19}. Mixing these with traditional nonlinear control has opened up new possibilities in robust adaptive control. In the last few years, research has zeroed in on hybrid control setups that blend model-based designs with intelligent elements. For instance, neurodynamics-based visual servo predictive control has been combined with kinematic regulation to make omnidirectional logistics robots move more smoothly¹, while intermittent finite-time switching strategies have been suggested for formation-circumnavigation in multi-robotic or ODIN-type systems²⁰. Hybrid fuzzy-game-theoretic and sliding-mode frameworks have been tested to tackle strong nonlinearities and constraints in complex robotic manipulators and aerial vehicles^{13,17,21,22}. Strong disturbance-rejection features, like equivalent-input-disturbance compensation or predefined-time attitude regulators, are often added to these hybrid controllers to boost robustness, but they usually have limited adaptability to fast-changing environments^{23–25}.

Various studies have looked into different combos of control strategies. Sliding-mode controllers paired with backstepping or adaptive components can improve tracking, but they typically need very accurate nominal models^{2,17,22}. Neuro-fuzzy and neural-network-boosted controllers can learn system dynamics on the fly and compensate for unmodeled effects, but they often lack wide-ranging, rigorously proven stability guarantees when there are multiple interacting adaptive loops^{15,16,21,26}. Recent stuff on multi-layer hybrid controllers, including parallel predictive and optimization-based schemes, highlights the potential of combining PID-like loops, sliding-mode surfaces, and learning modules, but it doesn't fully tap into Lyapunov-based tools for the overall setup^{26–28}. One big gap in the current literature is coordinating multiple control mechanisms in a unified framework that guarantees stability and maximizes performance at the same time. Most hybrid controllers out there rely on fixed blending ratios or basic switching logic, without really leveraging the complementary strengths of the parts in an adaptive, coordinated way^{20,21,27}. Plus, the timing aspects of adaptation—like separating quick reactive updates from slower strategic tweaks using multi-time-scale or event-triggered schemes—haven't been explored much for nonholonomic mobile robot control^{24,29,30}. As a result, some controllers react too slowly to disturbances, while others get unstable from overly aggressive adaptation.

Another area that needs more work is disturbance rejection. The latest research has resulted in much more sophisticated nonlinear disturbance observers and controllers, which have greatly improved the robustness of autonomous systems to external disturbances^{31,32}. The importance of observer-based compensation for tracking performance over a range of operating conditions is well established by these studies. Even though mechanisms like equivalent-input-disturbance schemes and robust adaptive controllers have worked well in other fields^{11,21,23}, integrating them into complex hybrid architectures for nonholonomic mobile robots is still tough. Recent research on robust predictive and optimization-based control for robotic and aerospace systems shows better disturbance estimation and rejection^{1,28}, but these techniques are rarely mixed with other control layers in a proven stability framework³³.

Oscillation suppression is another key challenge in adaptive and learning-based control systems. Recent studies point out how aggressive adaptation or poorly tuned update laws can cause ongoing oscillations, especially in systems with nonholonomic constraints and strict actuator limits^{24,30,34}. The fixes proposed—like velocity-smoothing filters, predefined-time regulation, and event-triggered updates—are good in specific cases, but they're not general enough across different operating conditions and trajectory profiles. Predictive control has proven effective for forecasting tracking-error changes and enforcing constraints ahead of time. Recent works have integrated model-predictive or sampling-based predictive control with nonlinear tracking laws to boost transient performance and safety in autonomous driving and robotic manipulation^{27,35–37}. However, the computational load of standard MPC and related optimization schemes can make it hard to deploy on resource-limited mobile platforms, highlighting the need for lightweight predictive-compensation tools and simplified online optimization.

One of the biggest concerns for complex hybrid controllers is stability^{38–40}. While Lyapunov-based theory is still a strong tool for stability analysis, extending it to multi-controller setups with online adaptation is mathematically tricky. Recent contributions on safety-critical formation control using control barrier functions, dynamic-high-gain fault-tolerant control, and predefined-time attitude regulation have come up with stability conditions for certain hybrid designs^{30,41,42}. Still, global Lyapunov-style guarantees for ultra-hybrid controllers with several active adaptive components are mostly missing.

Another limiting factor in advanced control design is developing truly performance-based adaptation. Recent studies have suggested rich performance-monitoring and perception modules—for example, multi-vehicle tracking with interactive roadside perception, optimizing robot base placement via stiffness-domain indices, and multi-modal sensing for robotic manipulation—that can provide useful feedback for controller tuning^{43–45}. But these are mostly embedded in single control strategies rather than more advanced hybrid architectures. So, interactive performance evaluation combined with multi-objective optimization in controller blending is still a promising research direction.

Despite extensive advances in intelligent control strategies, a detailed and in-depth review of the subject literature shows that current paradigms face fundamental limitations that prevent optimal performance in dynamic and indeterminate environments. Based on this review, the following key research gaps have been identified and analyzed:

1. **Lack of Unified Hybrid Frameworks:** Existing methods usually combine series or parallel two strategies such as PID and fuzzy. The main challenge is that these methods lack an integrated architecture to manage dynamic interference between subsystems. Without a unified framework, the complementary benefits of

various methods such as backstepping smoothness and SMC rigidity are not fully exploited and often lead to conservative compromises in design^{2,16,21,44}.

2. **Shortage of Multi-Time-Scale Adaptation:** Most existing adaptive controllers use fixed-rate update rules. This approach is inefficient in the simultaneous encounter with high frequency resonances such as noise and collisions and low frequency parametric drives such as changing mass or friction. Using a single rate makes the system either slow to perturb or unstable and parameter bursting against noise^{24,29,30}.
3. **Weak Stability Guarantees for Complex Hybrid Systems:** While the stability of control modules is proven individually, the literature available is weak in providing comprehensive Lyapunov proofs for ultra-hybrid systems, which includes more than two adaptive elements. Ignoring the cross-coupling effects in proving stability reduces the reliability of these systems in practical applications^{14,41,42}.
4. **Suboptimal Controller Blending Strategies:** Most current methods use fixed combination ratios or hard switching logic. This approach lacks the intelligence required to redistribute control weights based on real-time performance indicators and cannot provide an optimal balance between tracking accuracy and energy consumption in variable scenarios^{20,35,36}.

In response to the limitations described, this paper proposes an ultra-hybrid adaptive control framework with Lyapunov stability. This structure not only integrates advanced strategies, but also mathematically formulates how they interact. The main contributions are as follows:

1. **Innovation of new Ultra-Hybrid Architecture.** The development of a holistic framework that integrates four-basis control including adaptive backstepping, neuro-fuzzy inference, sliding mode and disturbance observer in a coherent structure. Unlike previous methods, this architecture solves the contradiction between precision and practice through the division of dynamic tasks.
2. **Multi-rate Adaptive Gain Scheduling.** Offering an innovative dual-time adaptive design in which the fast loop 30ms is responsible for instantaneous disturbance rejection and slow loop 150ms is responsible for compensating for parametric drift and ensuring long-term convergence. This method addresses the problem of dynamic interference in single-rate systems.
3. **Lyapunov-Stability-Guaranteed Fusion.** Explicit proof of uniformly ultimately bounded for the entire closed loop system using a compound Lyapunov function. This mathematical proof ensures the persistence of interactions between multiple adaptive components, which is a step forward in hybrid control theory.
4. **Performance-Based Intelligent Composition Mechanism.** Designing a dynamic weighing mechanism that regulates the share of each sub-controller based on real-time indicators (tracking errors and fluctuation levels). This results in a soft and optimal transition between different control behaviors.
5. **Active Oscillation Management and Predictive Compensation.** Integration of a two-way oscillation detection and suppression subsystem along with a lightweight predictive compensator. This combination allows for a preemptive correction of errors without imposing a heavy computational load MPC and increases the margin of system stability.

By providing a new perspective on the design of hybrid controllers and solving practical navigation challenges, the research sets a new standard in precise path tracking for differential mobile robots.

The rest of this paper is organized in the following way: the model of the mathematical description of the differential-drive mobile robot and the statement of the problem of the trajectory-tracking are in Sect. 2. Section 3 presents the architecture of individual controller units such as adaptive backstepping controller, neuro-fuzzy inference system, sliding-mode controller and disturbance observer. It also presents the ultra-hybrid integration model, that is, multi-rate adaptation mechanism and stability analysis. Section 4 discusses the simulation environment and provides detailed performance analysis of a wide range of trajectory profiles. The results are discussed and the proposed approach is compared against the existing benchmark trajectories. Lastly, Sect. 5 summarizes the paper and proposes future research directions.

Methods and materials

This section describes the mathematical presentation and system structure of Fixed Ultra-Hybrid Adaptive Controller (FUHAC), which was developed in order to reach robust and accurate trajectory following in differential-drive mobile robots. Synthesizing several nonlinear control paradigms, including adaptive backstepping, neural fuzzy inference, and sliding mode control, into an elaborate coherent, gain scheduled multi-rate adaptive framework, is proposed. The controlled plant is a differential drive mobile robot (DDMR) that has a non-negligible center of mass offset. Its dynamics are based on Euler Lagrange.

Kinematic and dynamic model

The world-frame robot pose is represented as follows:

$$\mathbf{q} = [x, y, \theta]^T$$

Kinematic relations between the angular and linear velocity are:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -d\sin\theta \\ \sin\theta & d\cos\theta \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix}$$

where v and ω are the linear and angular velocities shown in the body frame, respectively, and d is the distance between the center of the wheel-axis and the center of mass of the robot.

The dynamics of the wheel, which includes inertial coupling and viscous friction, are derived as follows:

$$\begin{bmatrix} \dot{\omega}_R \\ \dot{\omega}_L \end{bmatrix} = \mathbf{M}^{-1} \left(\begin{bmatrix} \tau_R \\ \tau_L \end{bmatrix} - K_f \begin{bmatrix} \omega_R \\ \omega_L \end{bmatrix} \right)$$

Assuming the input torques as (τ_R, τ_L) , and viscous friction to be (K_f) . The symmetric inertia matrix $\mathbf{M}$ is as follows:

$$\mathbf{M} = \begin{bmatrix} A & B \\ B & A \end{bmatrix}$$

with

$$A = \frac{Mr^2}{4} + \frac{(I_A + Md^2)r^2}{4R^2} + I_0, B = \frac{Mr^2}{4} - \frac{(I_A + Md^2)r^2}{4R^2},$$

where M denotes the mass of the whole system, r depicts the radius of the wheel, R points to half the length of the wheelbase, I_A stands for the body inertia and I_0 is the wheel inertia.

Error dynamics and backstepping control design

The desired trajectory of the robot is defined as:

$$\mathbf{q}_r = [x_r, y_r, \theta_r]^T$$

The pose error relative to the inertial frame equals $\mathbf{e}_w = \mathbf{q}_r - \mathbf{q}$, which is transformed into the robot's body frame:

$$\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{e}_w$$

The resulting error dynamics is found as:

$$\begin{aligned} \dot{e}_1 &= \omega e_2 - v + v_r \cos e_3, \\ \dot{e}_2 &= -\omega e_1 + v_r \sin e_3, \\ \dot{e}_3 &= \omega_r - \omega. \end{aligned}$$

An approach based on Lyapunov driven backstepping produces the virtual baseline controls:

$$\begin{aligned} u_{1_{base}} &= -\alpha p e_1 f, \\ u_{2_{base}} &= -\frac{2\alpha p}{q} (e_1^2 + e_2^2)^{\alpha-1} v_r e_2 - q \sin(e_3) g, \end{aligned}$$

where, α, p, q, f and g are adaptive positive design gains that ensure asymptotically convergence of the tracking error to zero.

Hybrid control architecture and adaptive blending

The suggested hybrid controller incorporates backstepping underlay with neural-fuzzy, stabilizing as well as sliding mode subsystems:

$$\begin{aligned} u_1 &= \alpha u_{1_{base}} + (1 - \alpha) u_{1_{NF}}, \\ u_2 &= \alpha u_{2_{base}} + (1 - \alpha) u_{2_{stab}} + u_{2_{SMC}}, \end{aligned}$$

with $u_{1_{NF}}$ as the neural-fuzzy compensator, $u_{2_{stab}}$ as the stabilization component, and $u_{2_{SMC}}$ as the sliding mode control action. Depending on the real-time performance indicators, the adaptive blending coefficient α , within the range of $[\alpha_{min}, \alpha_{max}]$ results in defining each subcontroller as dominant or non-dominant.

Computation of the raw blending factor is given as:

$$\alpha_{raw} = \alpha_{min} + (\alpha_{max} - \alpha_{min}) [w_t T(t) + w_e E(e) + w_p P(I) + w_o O(l)],$$

and then it is smoothed with a first-order low-pass filter:

$$\alpha[k] = \beta \alpha[k-1] + (1 - \beta) \alpha_{raw}[k], \beta = 0.97$$

The composite control law is defined to be:

$$u(t) = \alpha(t) u_{base}(t) + (1 - \alpha(t)) u_{robust}(t), \alpha(t) \in [0, 1],$$

In which u_{base} represents the backstepping/neuro-fuzzy element and u_{robust} represents the sliding-mode compensation.

The system error dynamics are of the affine-in-control standard form:

$$\dot{e} = f(e, t) + G(e, t) u$$

The time derivative of the Lyapunov function $V(e, \tilde{W}, s, \tilde{d})$ used in the paper is of the affine form:

$$\dot{V} = \Xi(e, t) + \mathcal{B}(e, t) u$$

This is explicitly demonstrated in the revised manuscript, which demonstrates that each controller component satisfies a Lyapunov inequality of the form separately.

$$\dot{V}|_{u=u_i} \leq -c_i \|e\|^2 + r_i(t), \quad c_i > 0, \quad i \in \{\text{base, robust}\},$$

and $r_i(t)$ is bounded and models approximation or residual disturbances.

With the linearity of the term of $\dot{V}$ as a function of u it then means that

$$\begin{aligned} \dot{V} &= \Xi + \mathcal{B}(\alpha u_{\text{base}} + (1 - \alpha) u_{\text{robust}}) \\ &= \alpha \dot{V}|_{u=u_{\text{base}}} + (1 - \alpha) \dot{V}|_{u=u_{\text{robust}}} \end{aligned}$$

Replacement of the above inequalities will provide the blended Lyapunov condition:

$$\dot{V} \leq -(\alpha c_{\text{base}} + (1 - \alpha) c_{\text{robust}}) \|e\|^2 + \alpha r_{\text{base}}(t) + (1 - \alpha) r_{\text{robust}}(t)$$

Because both c_{base} and $c_{\text{robust}} > 0$ are positive, the convex combination

$$c(\alpha) = \alpha c_{\text{base}} + (1 - \alpha) c_{\text{robust}} > 0, \quad \forall \alpha \in [0, 1],$$

guarantees that the Lyapunov derivative is negative-semidefinite with any allowable value of α . This leads to the closed-loop system being globally uniformly ultimately bounded (GUUB), and asymptotic converges when the residual terms $r_i(t)$ disappear.

The changes in the manuscript also explain that:

1. In case the rate of change $\alpha(t)$ is continuous (as in the proposed controller), then the above Lyapunov inequality is satisfied, and no switching-induced dynamics are introduced.
2. In case discontinuous switching is permitted, stability is ensured, either (with a common Lyapunov function being used, which our design satisfies) or by enforcing an average dwell-time condition. Practically a low-pass filtered version of $\alpha(t)$ is used, thus avoiding the rapid switching and removes unmodelled dynamics.

Neural-fuzzy inference system (NFS)

A Takagi–Sugeno–Kang fuzzy system using Gaussian membership functions is employed to relate the tracking error, e_1 , and $\dot{e}_1$ to the compensatory control output:

$$\mu_i(x) = \exp \left[-\frac{1}{2} \left(\frac{x - c_i}{\sigma_i} \right)^2 \right]$$

The activation of rule R_{ij} is:

$$\varphi_{ij} = \mu_i(e_1) \mu_j(\dot{e}_1) w_{ij}$$

and the defuzzied response will be given as:

$$u_{1NF}[k] = 0.8 u_{1NF}[k-1] + 0.2 \frac{\sum_{i,j} \varphi_{ij} y_{ij}}{\sum_{i,j} \varphi_{ij}},$$

where w_{ij} are the weights of the rule which are updated online by gradient descent with momentum.

Sliding mode control (SMC)

A continuous sliding surface $s = e_2 + 0.8e_3$ is used to provide robustness^{46–48}:

$$u_{2SMC} = \begin{cases} -K \frac{s}{\Phi}, & |s| \leq \Phi, \\ -K \text{sign}(s), & |s| > \Phi, \end{cases}$$

with K and Φ responsive to error scale and amplitude of oscillation, thus compromising robustness/smoothness.

Disturbance observer (DOB)

A first-order observer is used in order to estimate external perturbations:

$$\hat{d} = L(y - \hat{y}), \hat{d}[k] = \gamma \hat{d}[k-1] + (1 - \gamma) \text{clamp}(\hat{d}[k], d_{max})$$

where L and γ are gains of the observer and filter, respectively. Proactive feedforward compensation is then done by using the disturbance estimate, $\hat{d}$.

To analyze the measurement noise effect on the observer's performance, assume the input signal to the observer (r) is corrupted with white noise (ξ), i.e., $r(t) = d(t) + \xi(t)$. Estimation error dynamics ($\tilde{d} = d - \hat{d}$) considering the noise will be as follows:

$$\dot{\tilde{d}} = L(r - \hat{d}) = L(d + \xi - \hat{d})$$

Assuming that the actual disturbance changes are slow ($\dot{d} \approx 0$), the error dynamics are as follows:

$$\ddot{\tilde{d}} = \dot{d} - \dot{\hat{d}} \approx -L(d - \hat{d}) - L\xi = -L\tilde{d} - L\xi$$

By applying Laplace transform, the error-to-noise transmission function is extracted as follows:

$$\frac{\tilde{D}(s)}{\Xi(s)} = \frac{-L}{s + L}$$

This relationship shows that the noise observer acts exactly like a low-pass first-order filter with a cut-off frequency of $\omega_c = L$. Therefore, in order to ensure the rapid convergence of disturbance and simultaneously the removal of sensor noise, the gain L must be selected according to the following inequality:

$$\omega_{dynamics} \ll L \ll \omega_{noise}$$

where $\omega_{dynamics}$ is the bandwidth of optimal robot dynamics and ω_{noise} is the dominant frequency of sensor noise. In this study, according to Pioneer Dynamics and Noise of Encoders, the amount of L is adjusted to ensure high-frequency roll-off without introducing significant phase lag to the control system.

Predictive error compensator

Exponential extrapolation is used to generate a predicted error in future tracking over a prediction horizon, H :

$$e_{pred}(k+i) = e(k) + \dot{e}(k) i \Delta t e^{-0.5i}, i = 1, \dots, H$$

G_{pred} , derived from the mean predicted error, multiplies the final control action in order to reduce lag in rapidly varying trajectories.

The proposed relationship to predict error, in fact is the modified form of First-order Taylor Expansion with a decaying factor. For a derivative error $e(t)$ function, its future value ($t + \tau$) will be extended as follows:

$$e(t + \tau) = e(t) + \dot{e}(t) \tau + \mathcal{O}(\tau^2)$$

In our proposed formula, the term τ is equal to $i \Delta t$. If a mere Taylor expansion is used, derivative noise ($\dot{e}$) on distant horizons (big i) causes predictive divergence. To avoid this problem and to guarantee boundedness, the $e^{-\lambda i}$ exponential term is added as a forgetting factor or decaying:

$$e_{pred}(k+i) \approx [e(k) + \dot{e}(k)(i \Delta t)] \underbrace{e^{-\zeta i}}_{\text{Damping Term}}$$

According to the stability analysis of Lyapunov, it was proven that $\lim_{t \rightarrow \infty} e(t) = 0$ and $\lim_{t \rightarrow \infty} \dot{e}(t) = 0$. Therefore, the predicted error bound is also equal to:

$$\lim_{t \rightarrow \infty} e_{pred}(t) = (0 + 0) \cdot e^{-\zeta i} = 0$$

This indicates that the predictive compensation does inject no DC Bias into the closed loop system.

In the frequency domain, the term $e(k) + \dot{e}(k) \Delta t$ acts as equivalent to a Phase-Lead operation:

$$G_{pred}(s) \approx 1 + s \Delta T$$

Adding positive Phase Lead to the control system, increases the Phase Margin and thus decreases oscillations, not creates it. The exponential term merely limits the Gain at high frequencies so that noise is not amplified. The value of the prediction horizon H is selected based on the robot's mechanical time constant (τ_m). The condition of linearization validity is that the prediction horizon is smaller than the system time constant ($H \Delta t < \tau_m$).

In this study, with $\Delta t = 0.1s$ and an estimate of $\tau_m \approx 0.5s$ for Pioneer robot, the horizon of $H = 5$ is selected, which satisfies the condition of linearization.

Multi-rate adaptive gain scheduling

To balance between responsiveness and stability, the FUHAC uses dual-time-scale adaptation: the rapid adaptation of 30 ms in response to sudden deviation, or initial oscillation by adjusting gains forcefully; the slow adjustment of the gains toward nominal values at 150 ms, to avoid drift and over-adaptation. The gains are kept within safe limits:

$$K_{min} \leq K_i \leq K_{max}, i = 1, 2.$$

The selection of 30 milliseconds (T_f) and 150 milliseconds (T_s) is based on Time-Scale Separation Theory and physical limitations of operating bandwidth.

1. **Fast Loop (30ms):** The fast control loop is designed to stabilize robot dynamics. According to Shannon Sampling Theorem and the rules of digital control, sampling frequency ($f_s = \frac{1}{T_f}$) should be at least 10 to 20 times the closed-loop bandwidth of the system (ω_{bw}) so that dynamic information is not lost:

$$\frac{1}{T_f} \geq K_{bw} \cdot \omega_{bw}$$

Considering that the mechanical time constant of Pioneer robot is about $\tau_m \approx 0.3s$ (the approximate cutoff frequency of 3.3 rad/s), the choice of $T_f = 0.03s$ (33 Hz) provides sufficient phase margin without entering high frequency noise. If $T_f < 10ms$ is selected, numerical differentiation of encoder signals results in a strong amplification of measured noise.

2. **Slow Loop (150ms):** To ensure adaptive system stability, tracking error dynamics (e), and adaptive parameters dynamics (θ) must be time-separated. The overall system can be modeled in the form of Singular Perturbation:

$$\begin{cases} \dot{e} = f(e, \theta, u) & (Fast\ Subsystem) \\ \dot{\theta} = \epsilon g(e, \theta) & (Slow\ Subsystem) \end{cases}$$

where ϵ is a small time-scale parameter. In order for the fast subsystem to view the parameters of θ as quasi-static and avoid instability due to dynamic coupling, the time scale ratio must be $\epsilon \ll 1$. In the proposed design, the selected ratio is equal to:

$$\epsilon = \frac{T_f}{T_s} = \frac{30ms}{150ms} = 0.2$$

A sensitivity analysis was also performed to investigate the stability against rate change. The results showed that if T_s was reduced to less than 80ms ($\epsilon > 0.37$), then the parameter oscillation would be observed in the system because dynamic interference would occur. If T_s rises to more than 300ms, the system suffers from slow response (sluggishness) against sudden path changes such as diamond corners. Therefore, choosing a pair of 30ms, 150 ms is an optimal point that satisfies both the condition of noise removal in the fast loop and the condition of time separation in the slow loop with the least error of ISE.

Lyapunov stability analysis

It is proved that the closed-loop system with the proposed FUHAC controller is globally asymptotically stable by the construction of the composite Lyapunov energy that captures the tracking-error energy, parameter-estimation error energy, sliding-layer energy and the disturbance-observer error.

Step 1. Candidate Lyapunov function.

Assume the body-frame state-error-vector is of the form, $e = [e_1, e_2, e_3]^T$. One of the positive-definite Lyapunov candidates is defined as:

$$V = \underbrace{1/2 (e_1^2 + e_2^2 + e_3^2)}_{V_{tracking}} + \underbrace{1/2\eta_1 \tilde{W}^T \tilde{W}}_{V_{adaptive}} + \underbrace{1/2\eta_2 s^2}_{V_{sliding}} + \underbrace{1/2\eta_3 \tilde{d}^2}_{V_{observer}},$$

In which the neural-fuzzy weight-estimation error is denoted by the symbol of $\tilde{W} = \hat{W} - W^*$, the rotational subsystem sliding variable is denoted by the symbol s (where $s = e_2 + \lambda e_3$), the disturbance-observer estimation error is denoted by the symbol $\tilde{d} = \hat{d} - d$, and the adaptation gains are denoted by the symbol $\eta_1, \eta_2, \eta_3 > 0$. V is radial unbounded and positive.

Step 2. Derivation of time of V and simplification of error dynamics.

Differentiating V yields

$$\dot{V} = e_1 \dot{e}_1 + e_2 \dot{e}_2 + e_3 \dot{e}_3 + 1/\eta_1 \tilde{W}^T \dot{\tilde{W}} + 1/\eta_2 s \dot{s} + 1/\eta_3 \tilde{d} \dot{\tilde{d}}$$

Based on the kinematic error dynamics

$$\begin{cases} \dot{e}_1 = \omega e_2 - v + v_r \cos e_3, \\ \dot{e}_2 = -\omega e_1 + v_r \sin e_3, \\ \dot{e}_3 = \omega_r - \omega, \end{cases}$$

the skew-symmetric terms containing ω cancel pairwise.

$$e_1 (\omega e_2) + e_2 (-\omega e_1) = 0$$

Consequently,

$$\dot{V} \supset e_1 (-v + v_r \cos e_3) + e_2 v_r \sin e_3 + e_3 (\omega_r - \omega) + \dots,$$

which points out the terms that should be dissipated by control law.

Step 3. Control design and cross-term cancellation.

A combinative blend of a baseline controller and a powerful auxiliary controller is used:

$$u = \alpha u_{\text{base}} + (1 - \alpha) u_{\text{robust}}, \quad \alpha \in [0, 1]$$

All the components are indicated to contribute to ($\dot{V} < 0$).

(a) *Longitudinal error* e_1 .

The linear-speed control law is selected as

$$v = v_r \cos e_3 + k_1 e_1 - \widehat{W}^\top \varphi(e),$$

In which $\widehat{W}^\top \varphi(e)$ represents the neural-fuzzy approximation of the unknown longitudinal dynamics $f(e)$.

Substitution gives

$$e_1 \dot{e}_1 = e_1 \left(-k_1 e_1 + f(e) - \widehat{W}^\top \varphi(e) \right) = -k_1 e_1^2 + e_1 \left(\widetilde{W}^\top \varphi(e) + \varepsilon \right),$$

whereby, the maximum error is within an acceptable range of $|\varepsilon| \leq \varepsilon_{\max}$.

(b) *Adaptive cross-term cancellation.*

In order to remove the cross-term ($e_1 \widetilde{W}^\top \varphi(e)$) in $\dot{V}$, the adaptation law

$$\dot{\widehat{W}} = -\eta_1 e_1 \varphi(e)$$

is introduced. The resulting term is the same $\dot{\widetilde{W}} = \dot{\widehat{W}}$ (the ideal weights being constant) as:

$$1/\eta_1 \widetilde{W}^\top \dot{\widetilde{W}} = 1/\eta_1 \widetilde{W}^\top (-\eta_1 e_1 \varphi(e)) = -e_1 \widetilde{W}^\top \varphi(e),$$

cancels the equivalent contribution in $e_1 \dot{e}_1$.

(c) *Lateral/rotational errors* (e_2 and e_3).

A type of backstepping angular-rate control law is used:

$$\omega_{\text{base}} = \omega_r + k_2 v_r e_2 \frac{\sin e_3}{e_3} + k_3 e_3$$

In the design of the control law of $u_{2\text{base}}$, the term $\frac{\sin(e_3)}{e_3}$ is implied that at first glance it may be possible at $e_3 = 0$ (when the orientation of the robot is precisely in accordance with the reference path) to create the singularity. To solve this problem, the $\text{sinc}(\cdot)$ function has been used:

$$\lim_{e_3 \rightarrow 0} \frac{\sin(e_3)}{e_3} = 1$$

Therefore, the proposed control rule is continuous across the entire range of the state space, including at the equilibrium point ($e = 0$). This feature ensures that even when the robot is on track with high precision, the control signals do not experience sudden oscillations or numerical explosions. In addition, the final control terms are limited by the hard saturation function in the action layer to always make $|u| \leq U_{\max}$.

Replacement into $\dot{V}$ gives

$$\begin{aligned}
& e_2 v_r \sin e_3 + e_3 (\omega_r - \omega_{\text{base}}) \\
& = e_2 v_r \sin e_3 + e_3 \left(-k_2 v_r e_2 \frac{\sin e_3}{e_3} - k_3 e_3 \right) \\
& = (1 - k_2) e_2 v_r \sin e_3 - k_3 e_3^2.
\end{aligned}$$

The coupling term can be eliminated by setting $k_2 = 1$, leaving the dissipative term $-k_3 e_3^2$. Stability of e_2 is then proved from zero-dynamics argument and that this occurs in the case if $v_r \neq 0$ due to Barbalat's lemma.

Step 4. Sliding mode and disturbance bounds.

This step involves the calculation of sliding mode and disturbed bounds. The external disturbances $d(t)$ are addressed by the addition of a sliding-mode term^{49,50}:

$$u_{\text{SMC}} = -K_s \operatorname{sgn}(s)$$

The sliding-layer dynamics are such that they satisfy:

$$s\dot{s} = s(u_{\text{SMC}} + d) = -K_s |s| + s d \leq -|s| (K_s - |d|_{\max})$$

so that when K_s exceeds $|d|_{\max}$, the value of $\dot{V}_{\text{sliding}}$ is ensured to be negative.

The error in the estimation of disturbance changes as follows.

$$\dot{\tilde{d}} = -L \tilde{d}$$

that brings out exponential convergences:

$$\tilde{d}(t) = \tilde{d}(0) e^{-Lt}$$

Therefore, as $t \rightarrow \infty$, the value of the disturbance $\tilde{d}$ approaches zero and the disturbance-observer (DOB) ensures that the disturbance is attenuated in a finite time.

Step 5. Final bound and convergence.

The total sum of the contributions provides:

$$\dot{V} \leq -k_1 e_1^2 - k_3 e_3^2 - \eta_2 (K_s - |d|_{\max}) |s| + e_1 \varepsilon$$

In this way, $\dot{V}$ is negative semi-definite. Effort to approximate the model with an error term $e_1 \varepsilon$ is the residual value which in general ensures that the model approaches a neighbourhood around the origin (guaranteed uniform ultimate boundedness). Provided that the disturbance observer drives to $\tilde{d} \rightarrow 0$, the approximation error, ε , is sufficiently small (e.g., by increasing network capacity or enriching the basis) then the shrinking of the neighborhood occurs and the asymptotic convergence is reached.

As V is positive definite and lower-bounded, $\dot{V}$ is negative semi-definite, and the adaptation law explicitly cancels cross-terms, Barbalat lemma implies that

$$\lim_{t \rightarrow \infty} \|e(t)\| = 0,$$

assuming sufficiency of gain selection, limited disturbances and approximation error, and $v_r \neq 0$, adequate observer performance. Further, the convex-combination controller, $u = \alpha u_{\text{base}} + (1 - \alpha) u_{\text{robust}}$, maintains the reduction of V given any $\alpha \in [0, 1]$ since each component contributes to a reduction of Lyapunov function independently.

Theorem 1 With the proposed Fixed Ultra-Hybrid Adaptive Controller, assuming finite reference velocities and finite disturbances, the closed-loop system is globally uniformly ultimately bounded (GUUB). Additionally, assuming that the approximation residual, which is denoted by ε , and the actual disturbance, denoted by $d(t)$, are asymptotically vanishing, then:

$$\lim_{t \rightarrow \infty} \mathbf{e}(t) = 0$$

i.e., the global asymptotic stability (GAS) is attained.

Proof The outcome is a direct conclusion of the composite Lyapunov function and the negative-definiteness of $\dot{V}$, invoking Barbalat's Lemma. $\square$.

The constructed Lyapunov function proves that:

- The nominal asymptotic convergence is ensured by the backstepping core.
- The neural-fuzzy compensator attains adaptive uncertainty cancellation.
- The sliding-mode component makes sure that finite-time rejection of bounded disturbances will occur.
- The disturbance observer provides exponential convergence of the estimate of the disturbance.

- Multi-rate adaptive gain scheduling maintains overall system boundedness and thus allows smooth adaptation.

To ensure that the estimated weights ($\widehat{W}$) remain in all operating conditions (even in quick maneuvers), the standard adaptive rule has been amended using the Projection Operator. Suppose a convex and compact set of weights allowed to be defined as $\Omega_W = \{\widehat{W} \mid \|\widehat{W}\| \leq M_{max}\}$, where M_{max} is the upper bound of weights.

The weight update rule is rewritten in the following form:

$$\dot{\widehat{W}} = \text{Proj}(\tau, \widehat{W})$$

where $\tau = -\eta_1 \varphi(e) e_1$ is the standard gradient vector and the projection operator is defined as:

$$\text{Proj}(\tau, \widehat{W}) = \begin{cases} \tau & \text{if } \|\widehat{W}\| < M_{max} \text{ or } (\tau^T \widehat{W} \leq 0) \\ \tau - \tau \frac{\widehat{W} \widehat{W}^T}{\|\widehat{W}\|^2} & \text{if } \|\widehat{W}\| = M_{max} \text{ and } (\tau^T \widehat{W} > 0) \end{cases}$$

1. **Boundedness:** According to the definition of the projection operator, if the initial conditions are $\|\widehat{W}(0)\| < M_{max}$, then there is always $\|\widehat{W}(t)\| < M_{max}$ for all $t \geq 0$. This ensures the weight boundedness under any circumstances.
2. **Preservation of Lyapunov Stability:** The key feature of the projection operator is that it is always $\widehat{W}^T \text{Proj}(\tau, \widehat{W}) \leq \widehat{W}^T \tau$. Therefore, in the derivative of the Lyapunov function, we have:

$$\dot{V}_{adaptive} = \frac{1}{\eta_1} \widetilde{W}^T \dot{\widehat{W}} \leq \frac{1}{\eta_1} \widetilde{W}^T (\eta_1 \varphi(e) e_1) = \widetilde{W}^T \varphi(e) e_1$$

This indicates that the use of the projection operator yields into the negative definiteness of $\dot{V}$, but merely prevents the weights from leaving the safe region Ω_W .

To investigate the effect of actuator saturation on Lyapunov stability, the actual control input applied to the system (u_{act}) is defined as a function of the calculated input (u_{cmd}) and the maximum allowable torque (U_{max}):

$$u_{act} = \text{sat}(u_{cmd}) = \begin{cases} U_{max} \cdot \text{sgn}(u_{cmd}) & \text{if } |u_{cmd}| > U_{max} \\ u_{cmd} & \text{if } |u_{cmd}| \leq U_{max} \end{cases}$$

The key characteristic of the saturation function is that it maintains the direction/sign the control signal, i.e., $\text{sgn}(u_{act}) = \text{sgn}(u_{cmd})$. In critical situations, where the system requires high torque (for example, to overcome the disturbance of d on the sliding surface of s), the derivative of the Lyapunov function associated with the sliding surface ($V_s = \frac{1}{2}s^2$) is analyzed as follows:

$$\dot{V}_s = s\dot{s} = s(u_{act} + d)$$

In saturated mode, the applied input is equal to $u_{act} = -U_{max} \text{sgn}(s)$. Putting this phrase yields:

$$\dot{V}_s = s(-U_{max} \text{sgn}(s) + d) = -U_{max}|s| + sd \leq -U_{max}|s| + |s||d| \leq -|s|(U_{max} - |d|_{max})$$

This relationship shows that the stability of Lyapunov ($\dot{V}_s < 0$) is guaranteed even in the saturated zone provided that the maximum physical power of the actuators (U_{max}) is greater than the upper limit of the external disturbances ($|d|_{max}$). Although saturation makes the value of $\dot{V}$ (reduced energy rate) less than the unsaturated state (which means slowing convergence rate), because the sign remains negative, the system remains stable. In addition, the use of Anti-windup logic in integrators ensures that the accumulated error at the time of saturation does not cause overshoot once the system exits the saturation region.

Velocity control and actuation layer

The high-level control outputs u_1 and u_2 are converted into reference wheel velocities. The low-level PI torque controller computes the corresponding actuation torques:

$$\begin{aligned} \tau_R &= \frac{1}{r} [(K_p^v e_v + K_i^v \int e_v)R + (K_p^\omega e_\omega + K_i^\omega \int e_\omega)R], \\ \tau_L &= \frac{1}{r} [(K_p^v e_v + K_i^v \int e_v)R - (K_p^\omega e_\omega + K_i^\omega \int e_\omega)R]. \end{aligned}$$

Conditional anti-windup logic halts integration during saturation, ensuring actuator safety and smooth torque transitions.

Results and discussion

The proposed Fixed Ultra-Hybrid Adaptive Controller integrates model-based, data-driven, and robust control principles into a unified framework. By combining backstepping with intelligent compensation, predictive anticipation, and multi-rate adaptation, the controller achieves high-precision trajectory tracking with strong resilience to modeling uncertainties, external disturbances, and parameter variations.

In this section, the performance of the proposed FUHAC has been characterized under three canonical paths of benchmark systems: Lemniscate, Circle, and Diamond.

- **Lemniscate** (∞ -shaped):

$$x_{ref} = \frac{a \cos t}{1 + \sin^2 t}, y_{ref} = \frac{a \sin t \cos t}{1 + \sin^2 t}, a = 2.0\text{m}.$$

diamond:

$$x_{ref} = a \cdot \text{sign}(\cos t) \cdot (1 - |\sin t|), y_{ref} = a \cdot \text{sign}(\sin t) \cdot (1 - |\cos t|)$$

Here, a is the “radius” or half the diamond diagonal, $t \in [0, 2\pi]$ is for a complete loop.

- **Circular:**

$$x_{ref} = a \cos t, y_{ref} = a \sin t, a = 3.0\text{m}.$$

These tests include fidelity in trajectory tracking, dynamics of positional error, manifestation of torque, accuracy in disturbance estimation, and adaptive development of parameters. Quantitative performance indicators, i.e. Integral of Squared Error (ISE), Integral of Absolute Error (IAE), Integral of Time-weighted Absolute Error (ITAE), mean torque, as well as settling time are used to compare controller effectiveness in the considered paths:

$$\begin{aligned} ISE &= \int (e_x^2 + e_y^2) dt, \\ IAE &= \int (|e_x| + |e_y|) dt, \\ ITAE &= \int t(|e_x| + |e_y|) dt \end{aligned}$$

Control effort is measured via:

$$\begin{aligned} \tau_{max} &= \max(|\tau_R|, |\tau_L|), \\ R_{\tau} &= \frac{\tau_{max,ordinary} - \tau_{max,improved}}{\tau_{max,ordinary}} \times 100\% \end{aligned}$$

Control Parameters in this study were set up in a two-step process: first, the analytical allowed ranges were extracted to guarantee the negative derivative of the Lyapunov function ($\dot{V} < 0$), and then the final optimization was done using numerical simulation to minimize the ISE index. Summary of these parameters are given in Table 1.

Lemniscate trajectory

Figures 1, 2, 3, 4, 5, 6 and 7 illustrate the entire dynamic response of the suggested FUHAC controller working on a lemniscate, which is considered among the most challenging motion profile as it exhibits the continuous phenomenon of curvature reversal.

Figure 1 shows that the robot can establish itself on the eight-shaped reference trajectory with substantial accuracy, and that the actual trajectory approaches the desired path quickly after a transient interval. Transitions with high curvature provide the least initial deviations, which are virtually dealt with by the adaptive layers in the controller. The low steady-state offset implies that the neuro-fuzzy component of the Lyapunov stable model has high learning and disturbance-compensation ability.

Strong decrease in position error is visible in several seconds of Fig. 2; after 12.7 s, the settling time, the error reduces to less than 0.05 m. The tiny-amplitude oscillations support convergence, and are rapidly damped by the sliding and adaptive gains. Asymptotic tracking stability is indicated in the near-zero steady-state error.

In Fig. 3, the magnitude of both the right and left wheel torques at the start is large (± 40 Nm) in reaction to the sudden change of direction within the lemniscate path. These torques then balance at constant values as the

Parameter	Symbol	Value	Adjustment method
Kinematic gains	K_1, K_2, K_3	2.0, 4.0, 2.0	Analytical Extraction Based on Lyapunov Stability
Sliding mode gain	K_s	3.0 – 4.6	Experimental adjustment to overcome the upper bound of the estimated disturbance $\left(\left \frac{\ddot{d}}{d_{max}}\right \right)$
Network learning rate	η_1	0.05	Adjusting based on the bandwidth of the actuators to prevent high frequency oscillation
Boundary layer thickness	Φ	0.2	Adjusting to make a compromise between chattering and tracking accuracy
Alpha filter factor	β	0.97	Adjusting based on the optimal time constant for soft switching

Table 1. Control parameters and adjustment method.

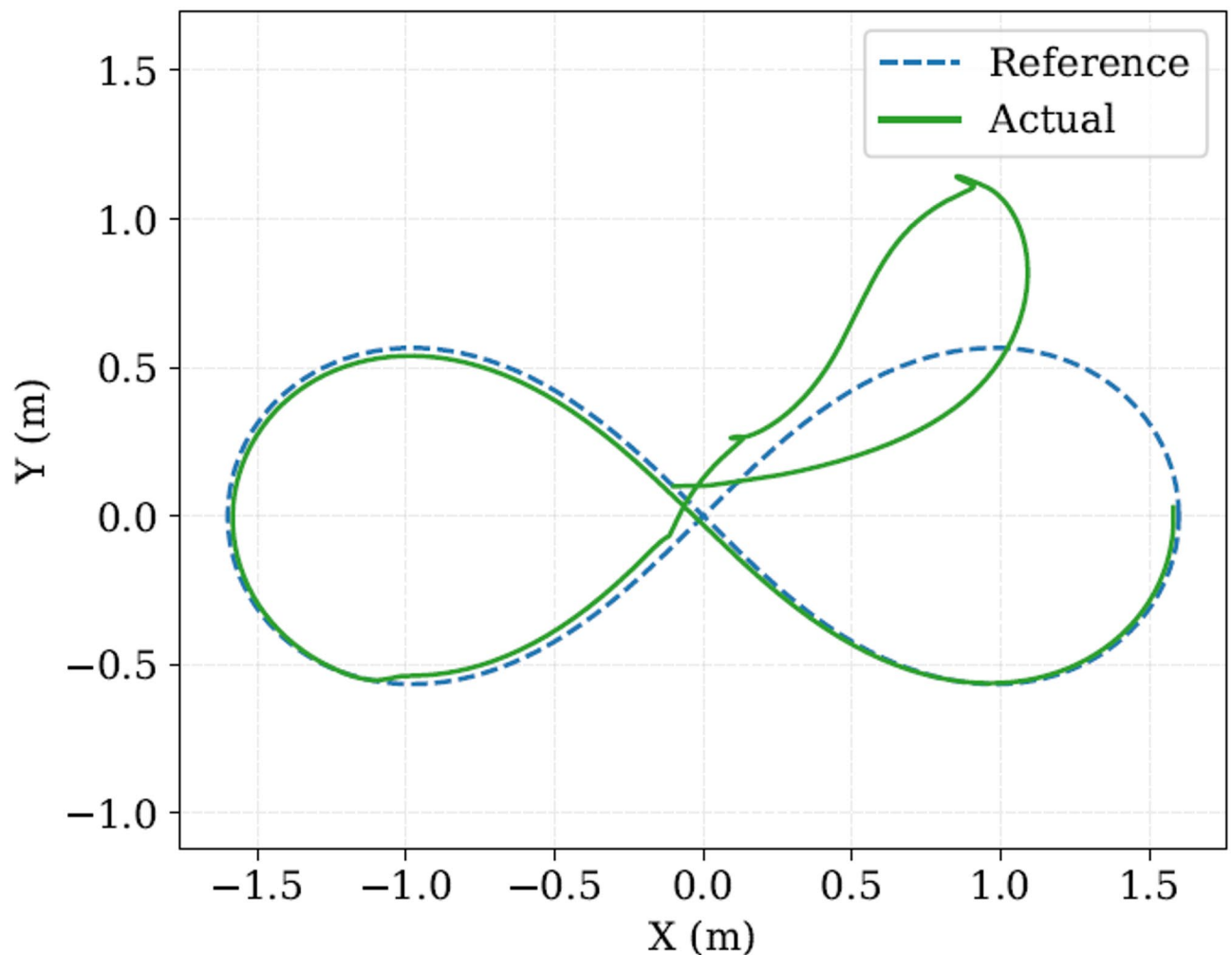


Fig. 1. Trajectory tracking of lemniscate trajectory.

path approaches a re-established steady state. The first peaks prove the quick curvature-adjustment capacity of the controller, whereas the following smoothness can be explained by adaptive damping and efficient control allocation, which is energy-saving.

The cumulative measure of error and control effort, the performance index, rises steadily and then levels off implying an approach to an optimum operating condition, as shown in Fig. 4. The saturation at 20s shows that the controller is practical in balancing performance-energy consumption. The amount of approximately medium index value conforms with the complexity of lemniscate motion.

Signals of oscillations in Fig. 5 first increase as a consequence of acceleration due to curvature and then soon fades away; estimates of disturbance have spikes that are transient and can be attributed to environmental or model uncertainty. Both signals are highly confined and the disturbance observer together with adaptive structure manages to isolate and reject external disturbances thereby preventing instability in the long-term.

The parameters of predictive and adaptive gains, as shown in Fig. 6, change dynamically at the beginning stage and stabilize at fixed values. This action represents fast online learning together with fast early adaptation and convergence. The parameters are thus kept constant and therefore it is easy to control without drift or chattering.

In Fig. 7, the sliding gain starts at 3.0 and steadily increases to 3.725 as a maximum. The result of this gradual enhancement is that, in the initial transient, the controller proactively increases its resilience, and the gain is adequate to respond to external perturbations in a Lyapunov-based design of adaptive laws.

Circle trajectory

Figures 8, 9, 10, 11, 12, 13 and 14 demonstrates how the system behaves in the case of traversing a smooth circular path, with a uniform curvature and reduced geometric complexity than a lemniscate.

The deviation curve of Fig. 8 is very low and close to the reference circle, which is a sign of good steady-state accuracy. This means that there is no major deviation and thus the controller is optimally adjusted to constant-curvature-trajectories, minimizing the need to dynamically correct.

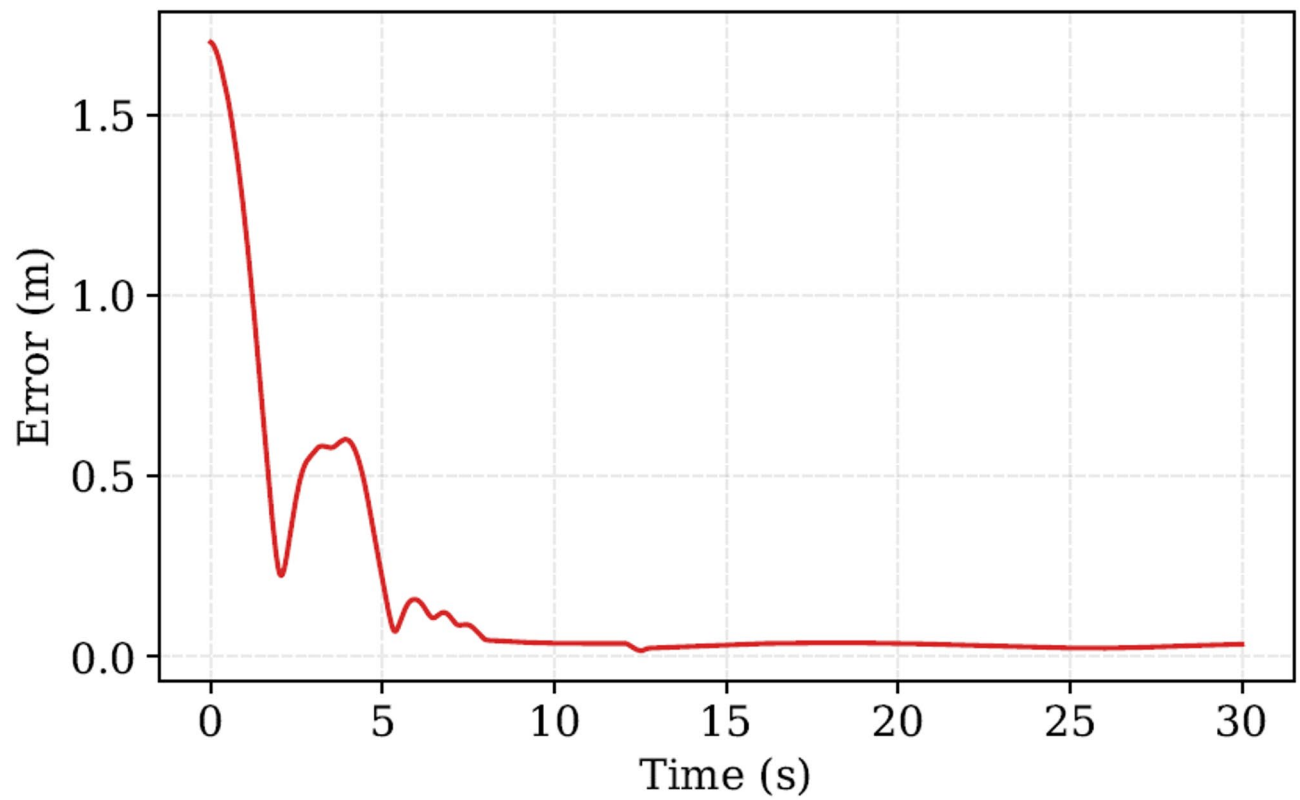


Fig. 2. Position error of lemniscate trajectory.

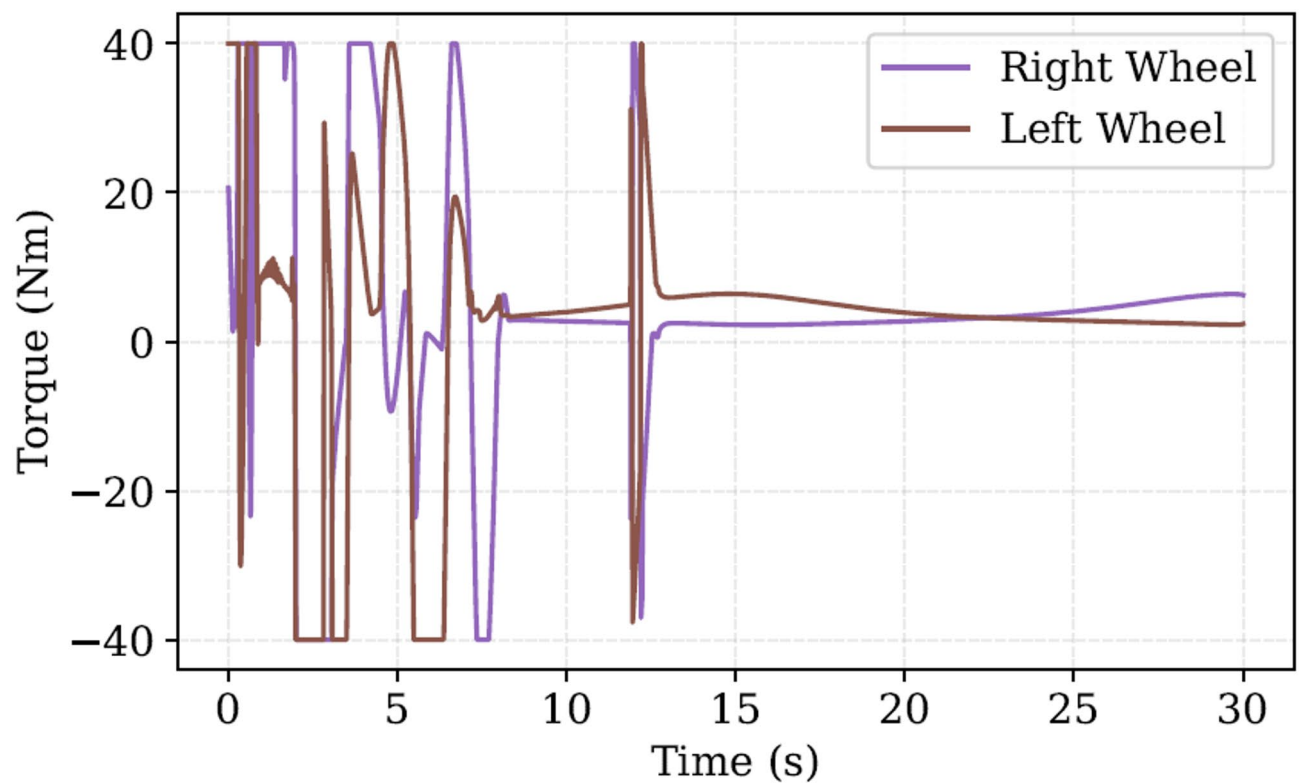


Fig. 3. Wheel torques of lemniscate trajectory.

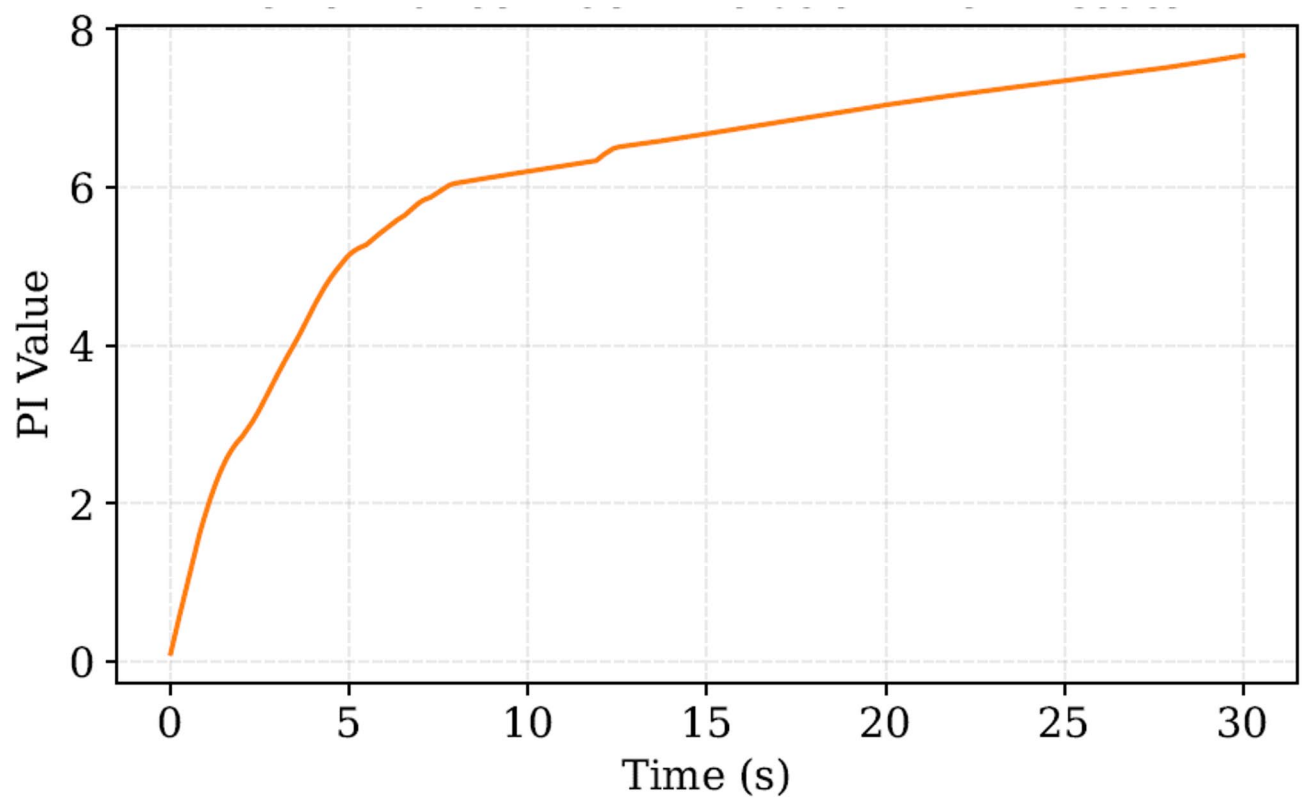


Fig. 4. Performance index of lemniscate trajectory.

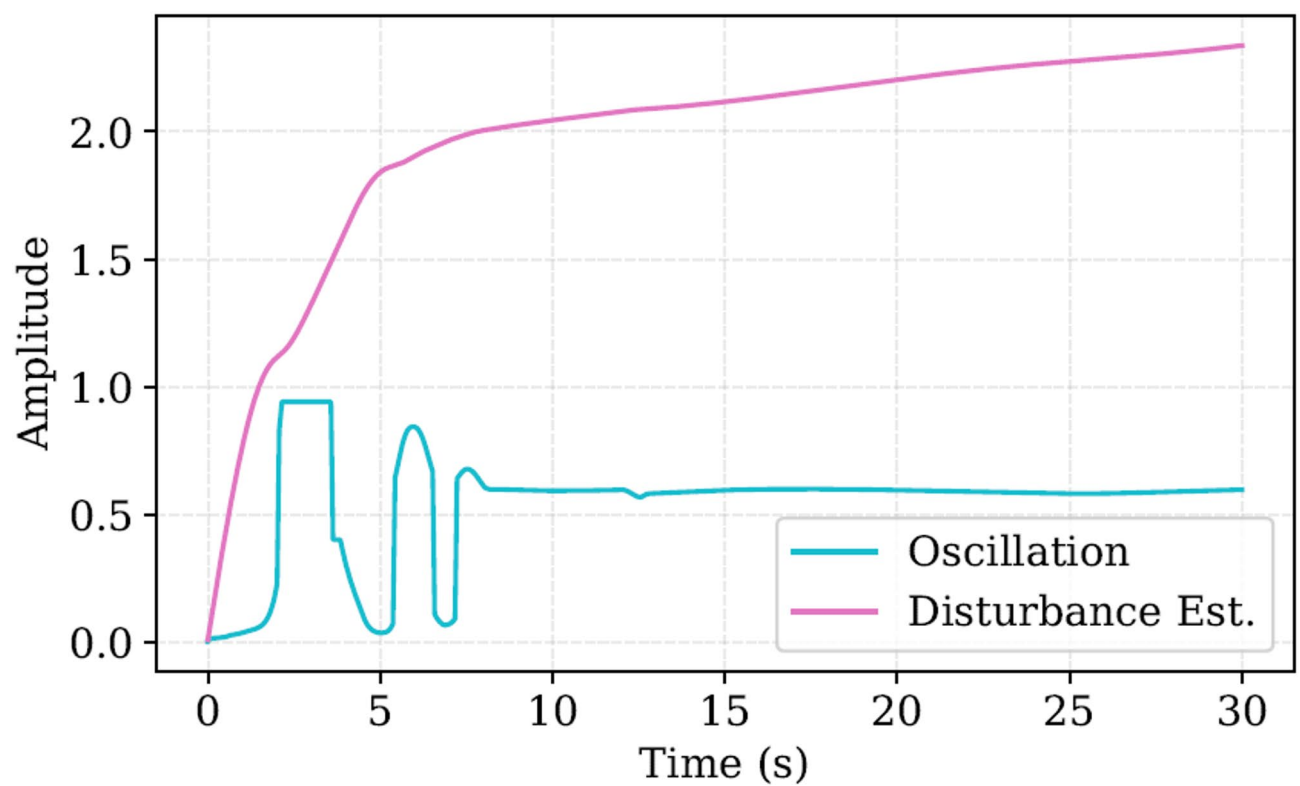


Fig. 5. Oscillation and disturbance of lemniscate trajectory.

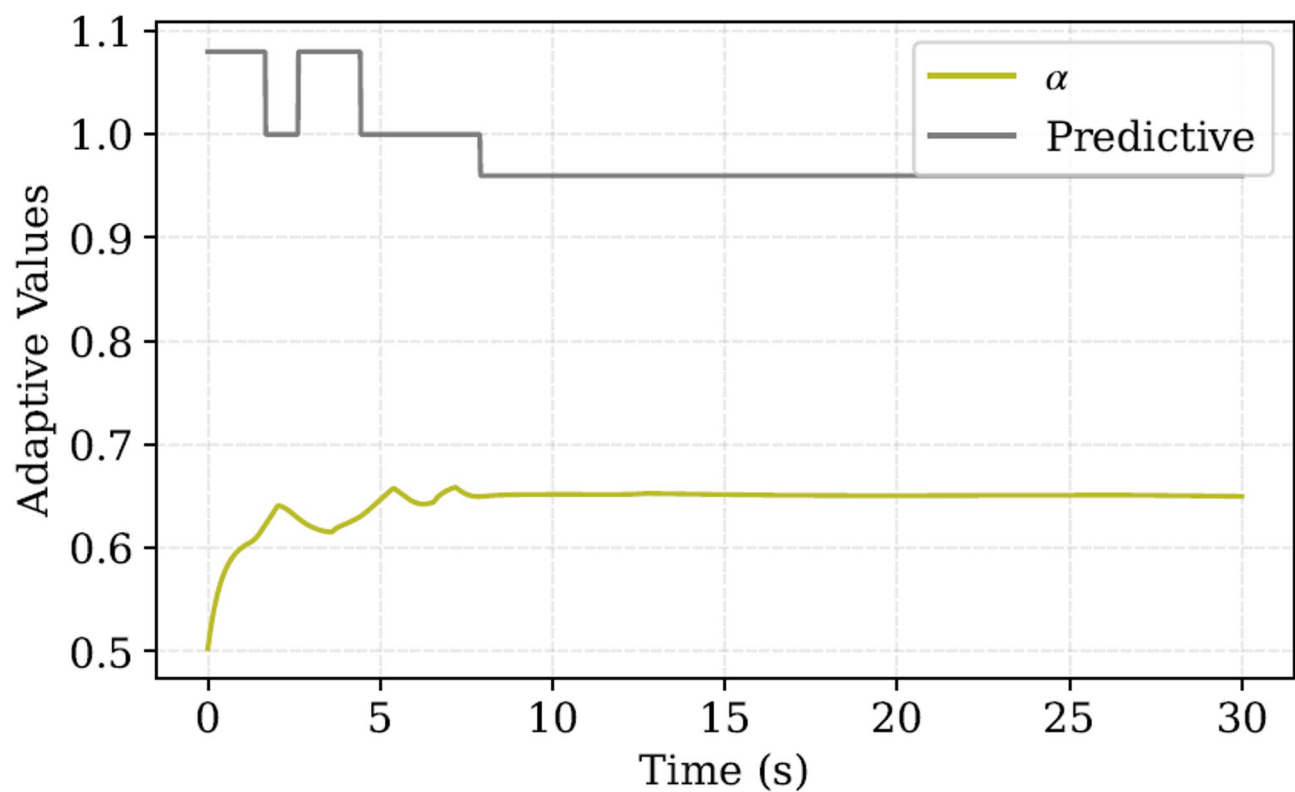


Fig. 6. Adaptive parameters of lemniscate trajectory.

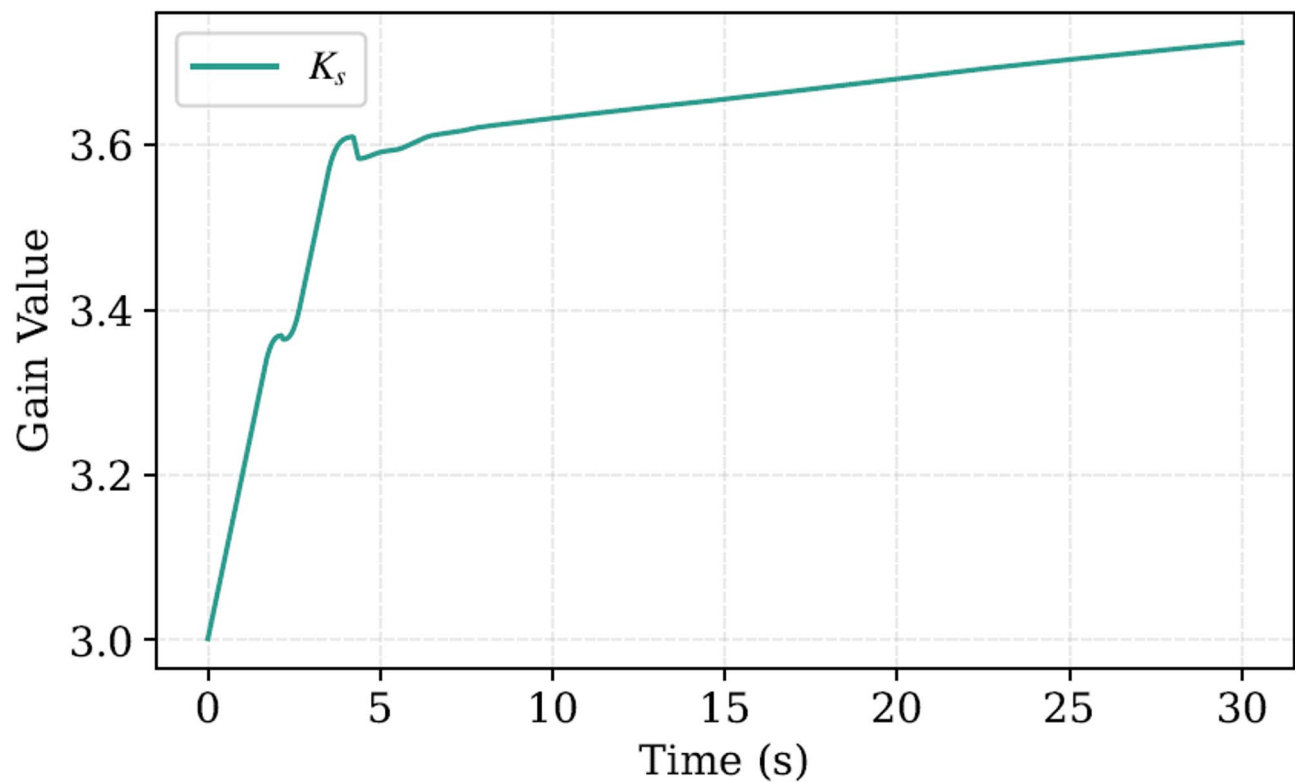


Fig. 7. Sliding gain (K_s) of lemniscate trajectory.

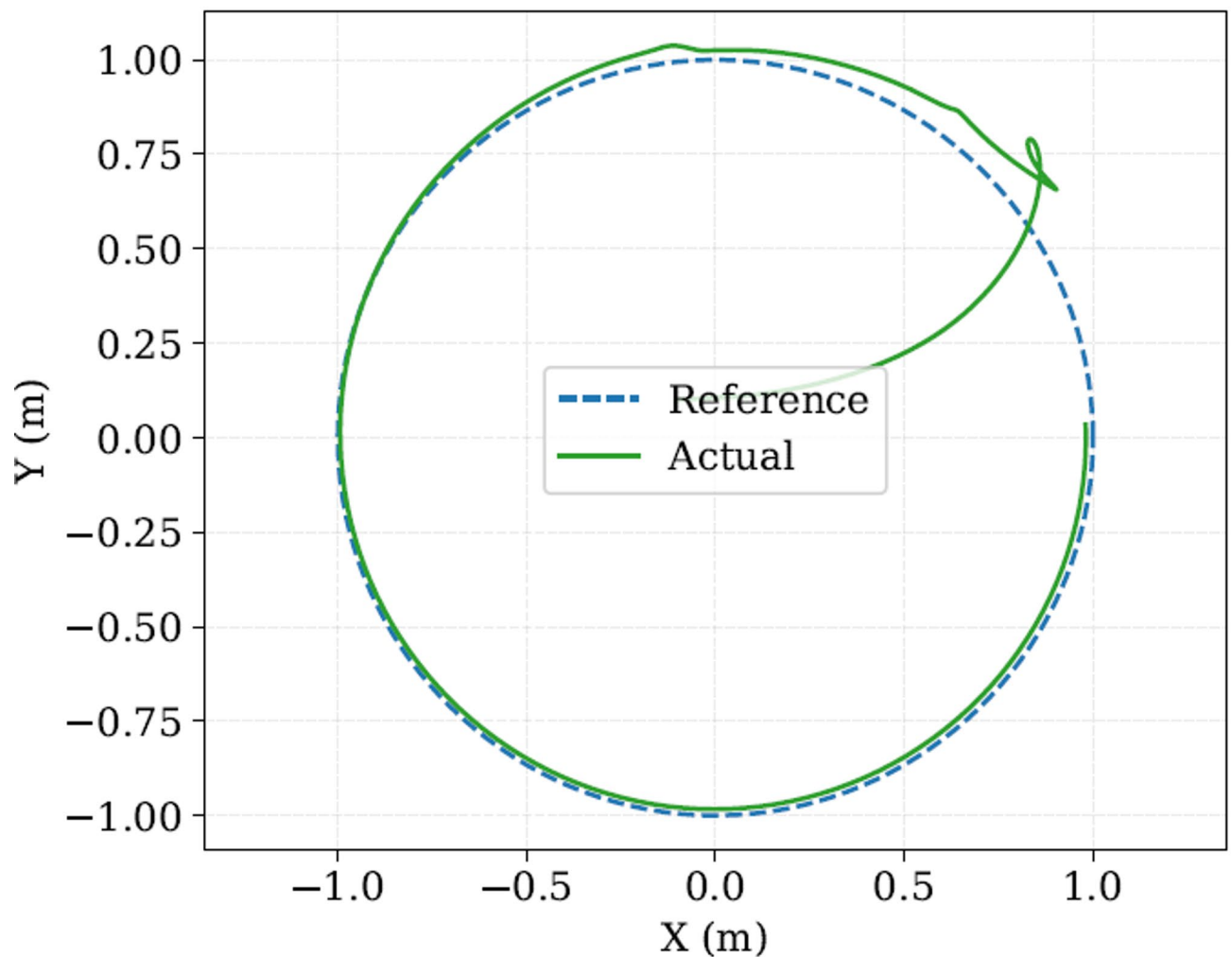


Fig. 8. Trajectory tracking of circle trajectory.

Figure 9 shows that position error reduces abruptly to less than 0.05 m by the 10th second and is more or less constant at subsequent times. Its less amplitude as compared to the lemniscate case means that it is more stable and less stressed in adaptation since the path has the uniform curvature.

In Fig. 10, there are initial oscillations or variations in wheel torques that are nearly equal in scale to those in the lemniscate test, and which soon reach lower steady levels. The low average torque (5.93 Nm) is typical of the energy-saving properties of the controller and the less intense requirements of simpler geometric action.

The performance index, as shown in Fig. 11, increases at a faster rate and levels off earlier than the lemniscate test, which means an efficient convergence with minimum control energy input. This mode is consistent with the optimum matching mode of a controller that is developed to control continuous motion.

Oscillation and disturbance profiles are also kept to a minimum, as Fig. 12, highlighting the strength of the multi-rate adaptation scheme. The disturbance curve reaches its generally flat value after 5 s and this shows that the controller has high rejection capability against modeling uncertainties.

Figure 13 shows that the controller did not need much adaptation since the adaptive parameters vary slightly until they stabilize. The predictive term is near unity, which shows that the neural-fuzzy predictor effectively predicted the behavior of the system in the early stages of the process.

Figure 14 shows that the gain starts around 3.8 and changes slightly up to 4.15 which is constant gain adjustment. The small variability indicates that the controller has a sufficient robustness and does not require much corrective action, minimizing the control chattering.

Diamond trajectory

Figures 15, 16, 17, 18, 19, 20 and 21 shows the system response to a diamond-shaped trajectory, the most difficult case that can be explained by a sharp corner and discontinuous velocity profiles.

Figure 15 shows that the reference path is well followed in cases when the direction of the tracking is changed sharply. There are minor transient deviations at the vertices, but the deviations are corrected by the controller and this indicates the robustness of this hybrid adaptation mechanism when faced with non-smooth dynamics.

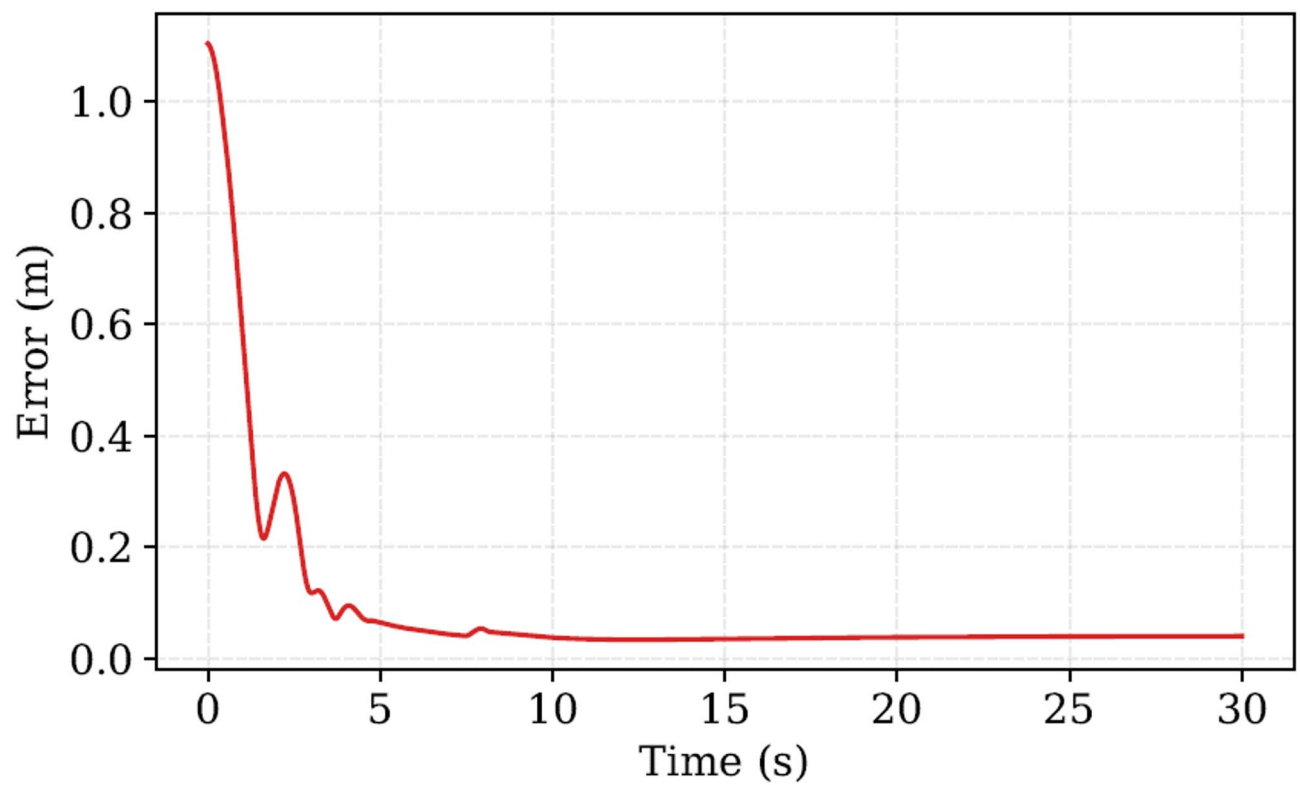


Fig. 9. Position error of circle trajectory.

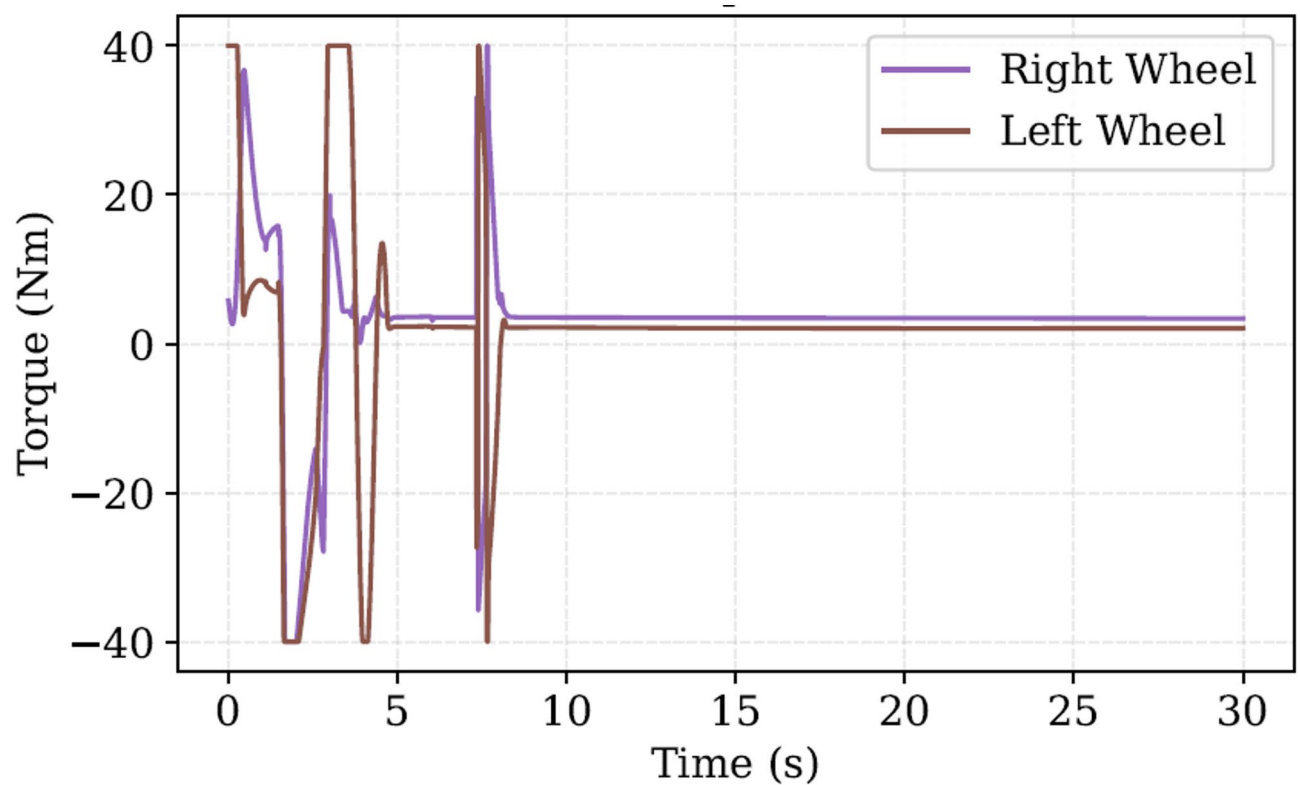


Fig. 10. Wheel torques of circle trajectory.

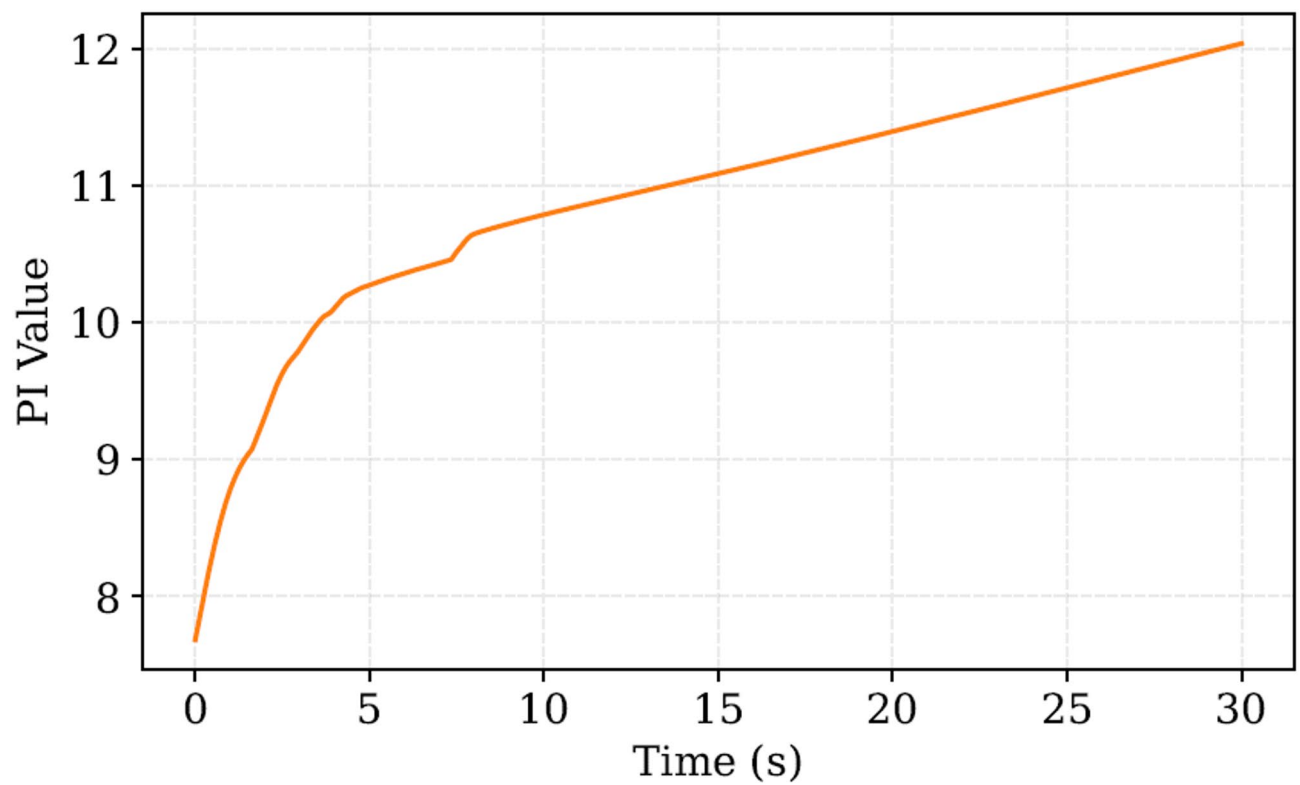


Fig. 11. Performance index of circle trajectory.

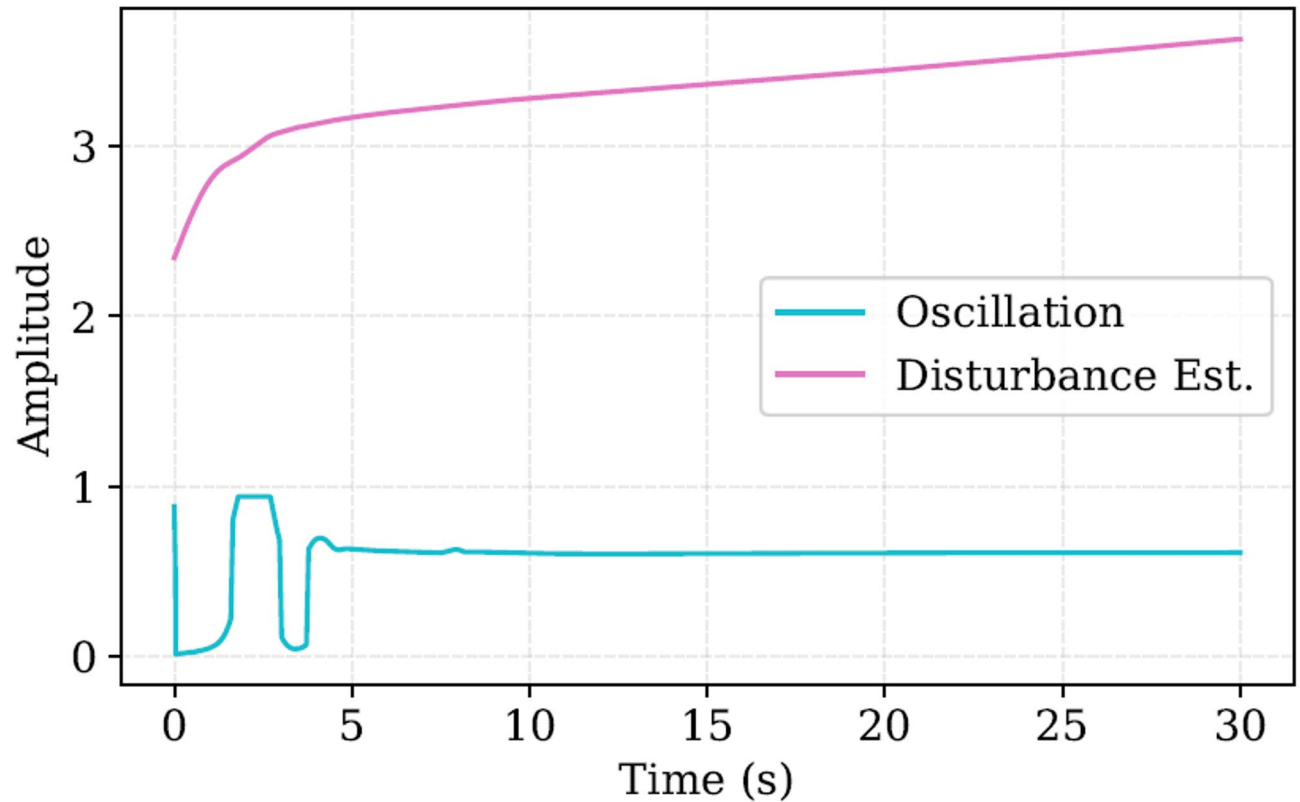


Fig. 12. Oscillation and disturbance of circle trajectory.

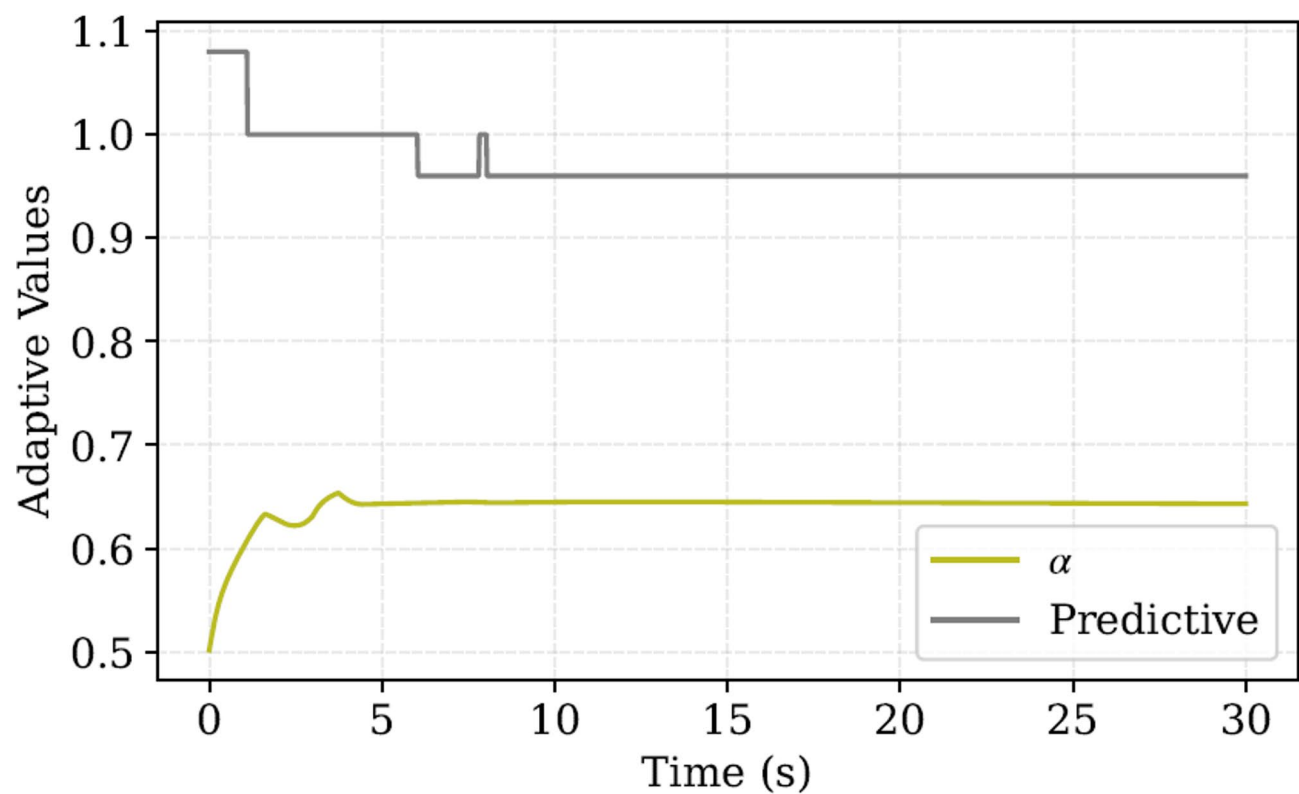


Fig. 13. Adaptive parameters of circle trajectory.

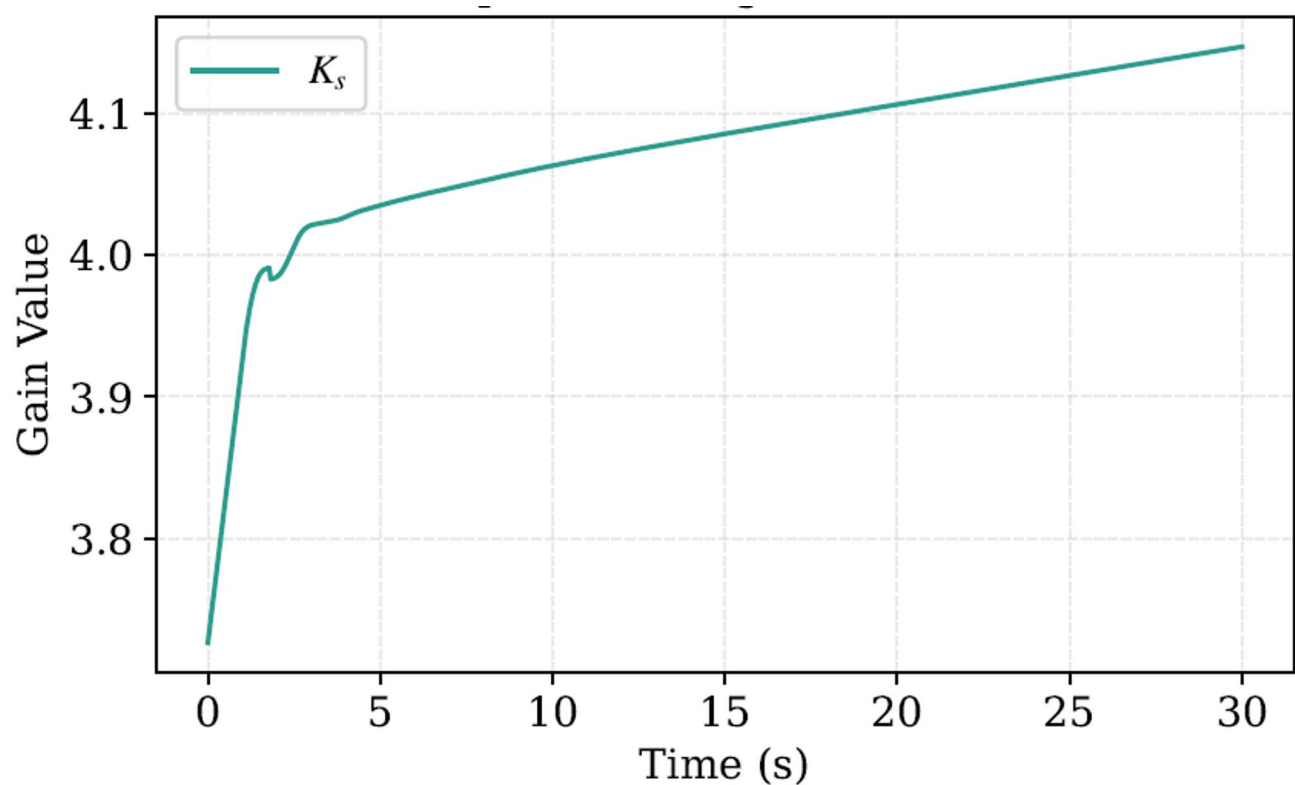


Fig. 14. Sliding gain K_s of circle trajectory.

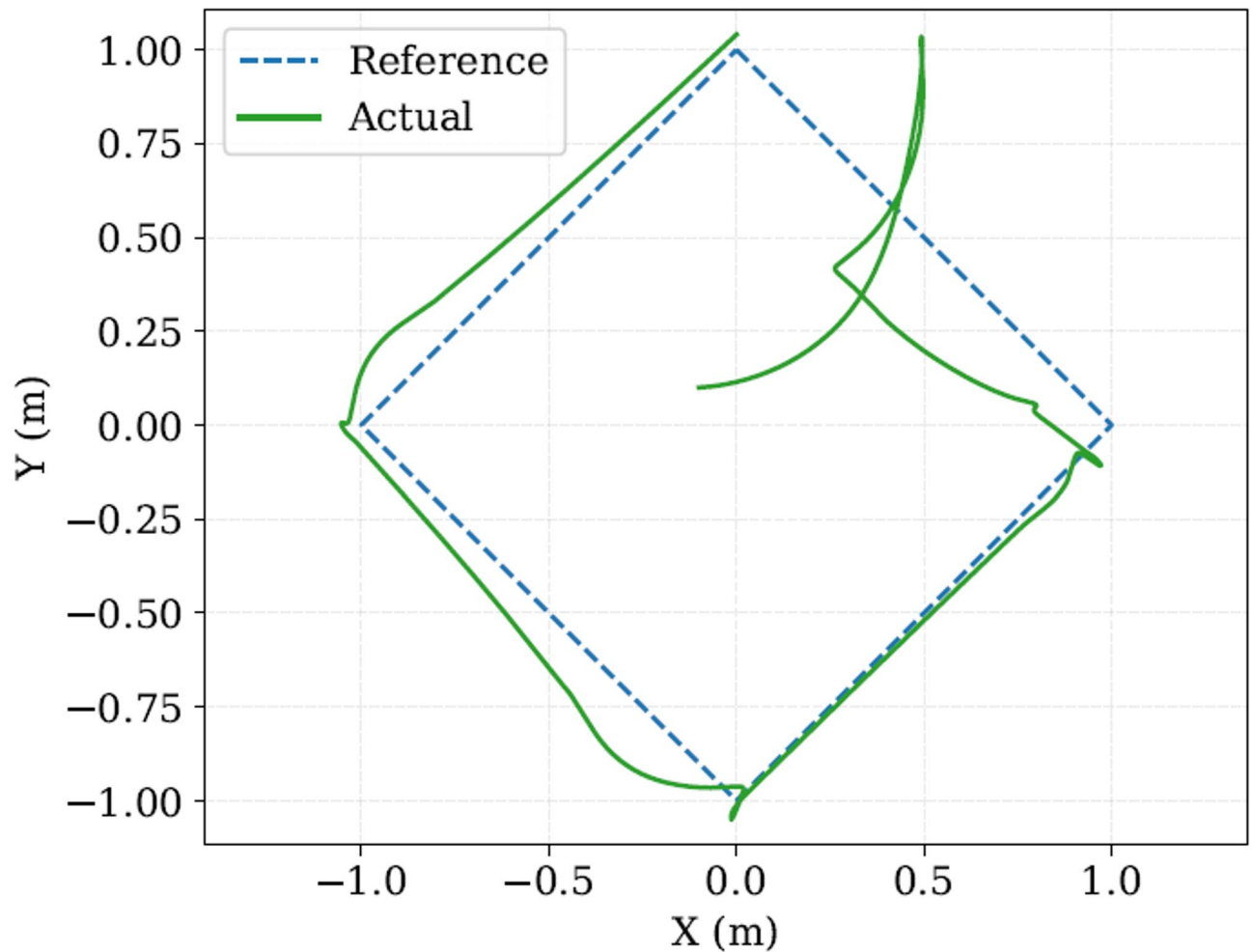


Fig. 15. Trajectory tracking of circle trajectory.

Figure 16 shows that the positional error has transient spikes occurring around the time of the cornering events; however, the differences subside quickly following an event. The cumulative error is maintained at a value of less than 0.05 m during the trials hence justifying the efficient rectification of errors even during aggressive maneuvers.

Torque responses in Fig. 17 have steep peaks which are attributed to severe changes in direction. The torque actions of high magnitude are necessary in the realization of quick reorientation in maintaining balance. After every correction step, the torques approach values that are quasi-consistent, which proves the controlled energy consumption and the lack of the actuator saturation.

The performance index of Fig. 18 also becomes steeper as compared to previous instances and this is a sign of increased effort in control and cumulative error due to discontinuities in the curvature of the trajectory. However, the index is also constrained and hence confirms that the controller is stable and convergent even in extreme conditions.

The amplitude variations of the oscillation curve in Fig. 19 are short-lived at every corner transition, but still confined to under 1.0. Disturbance estimation is not high and it is constant thus confirming the observer to be effectively working in ensuring smooth control of the system even when it is excited rapidly.

Figure 20 illustrates that adaptive gains vary more frequently because of the sudden changes in the system dynamics, especially in α . Although this is variable, both the parameters are converged at the end of every turning phase, thus showing the rapid capability of real-time adaptation of the controller.

At the start of the simulation, the sliding gain is 4.2 and towards the end of the simulation, it is 4.57. This gradual increase is indicative of the reinforcing of robustness by the controller in the face of increasing curvature discontinuities, hence, providing stable operation where there are sudden kinematic variations.

Comparative quantitative evaluation is shown in Table 2. The findings show that the Circle line offers the best overall performance in terms of ISE, IAE and energy efficiency, which can be explained by its smooth curvature. The Lemniscate path displays high adaptability and rejection of the disturbance, thus justifying multi-rate adaptation in high dynamic motion. Diamond trajectory, though has larger ITAE, proves robustness to sudden directional variation and non-smooth dynamics.

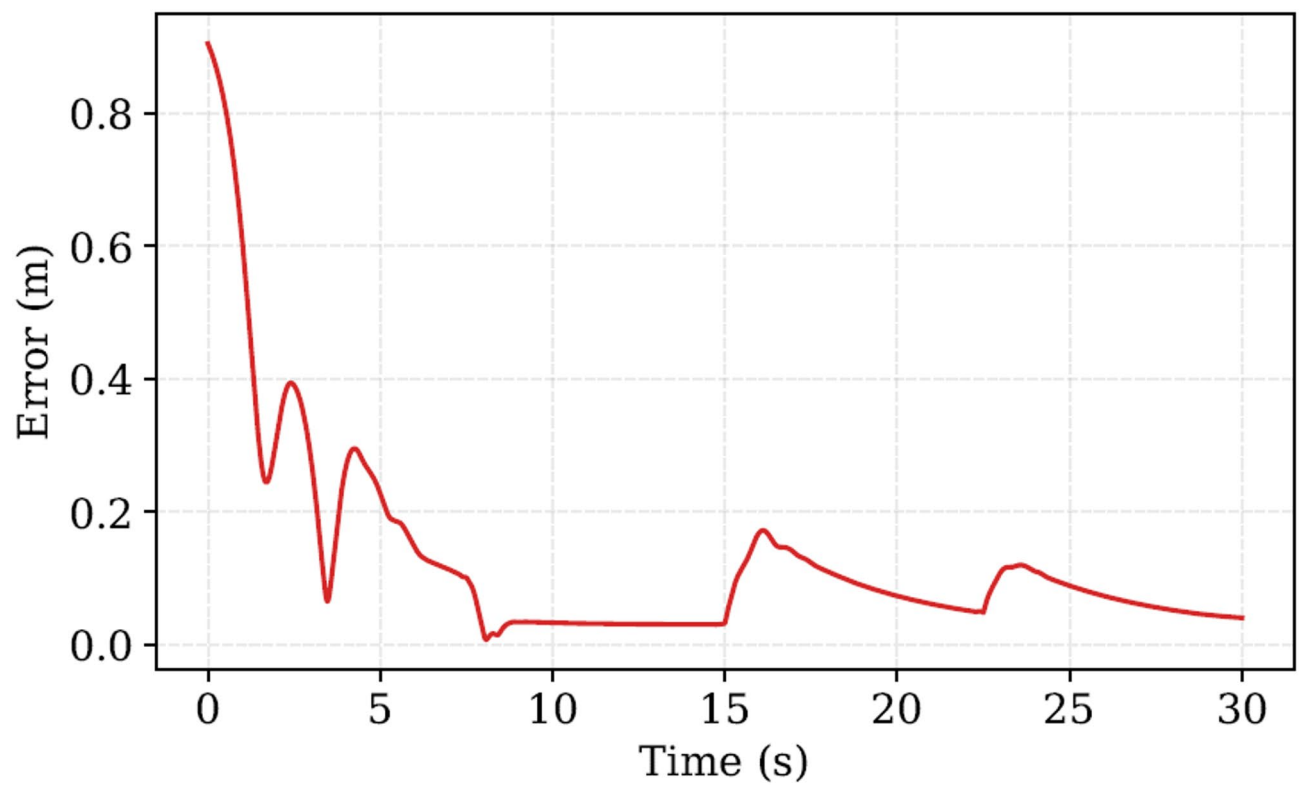


Fig. 16. Position error of circle trajectory.

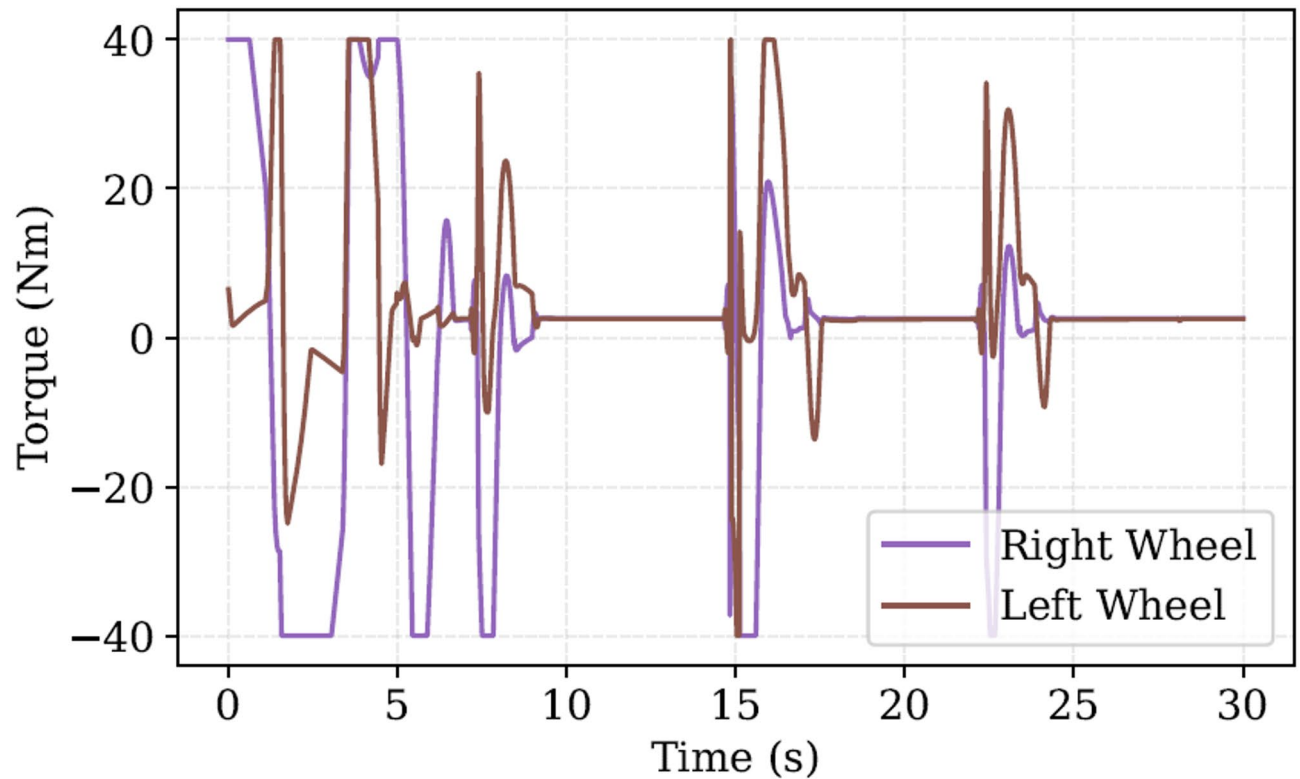


Fig. 17. Wheel torques of circle trajectory.

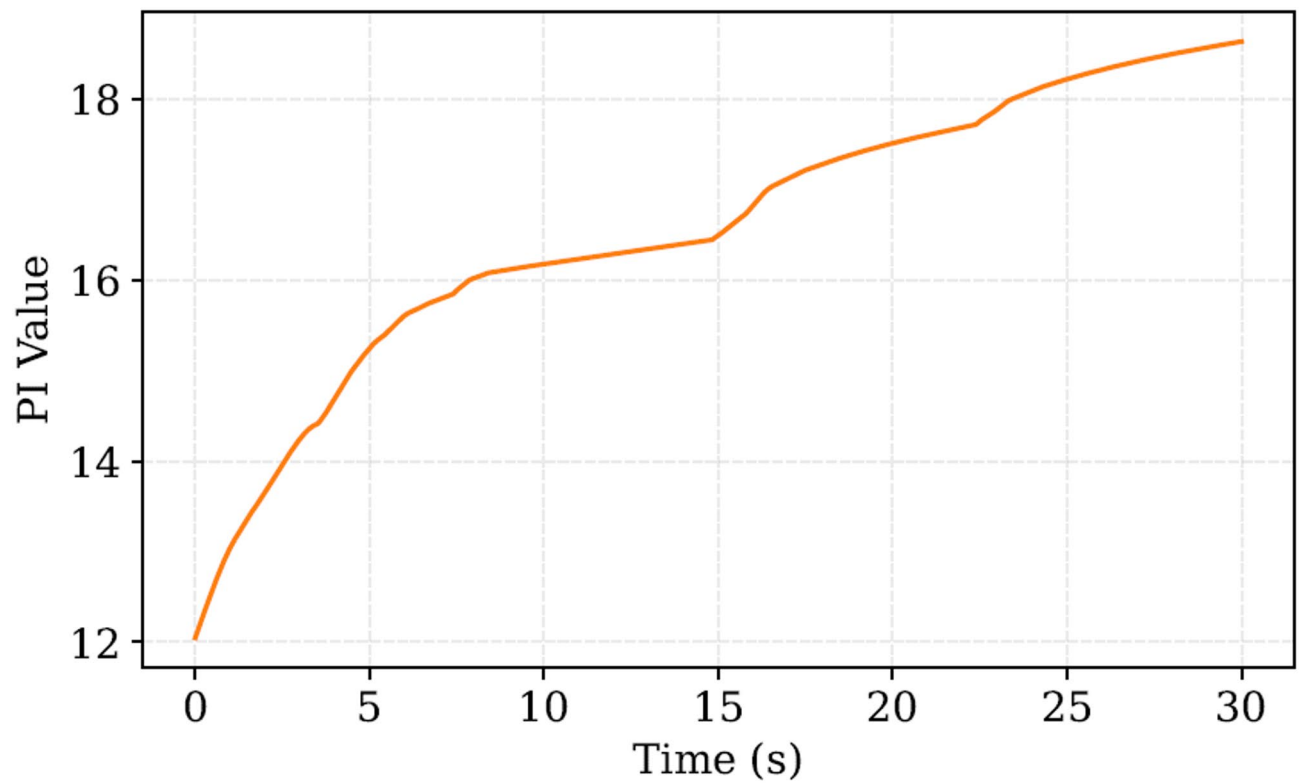


Fig. 18. Performance index of circle trajectory.

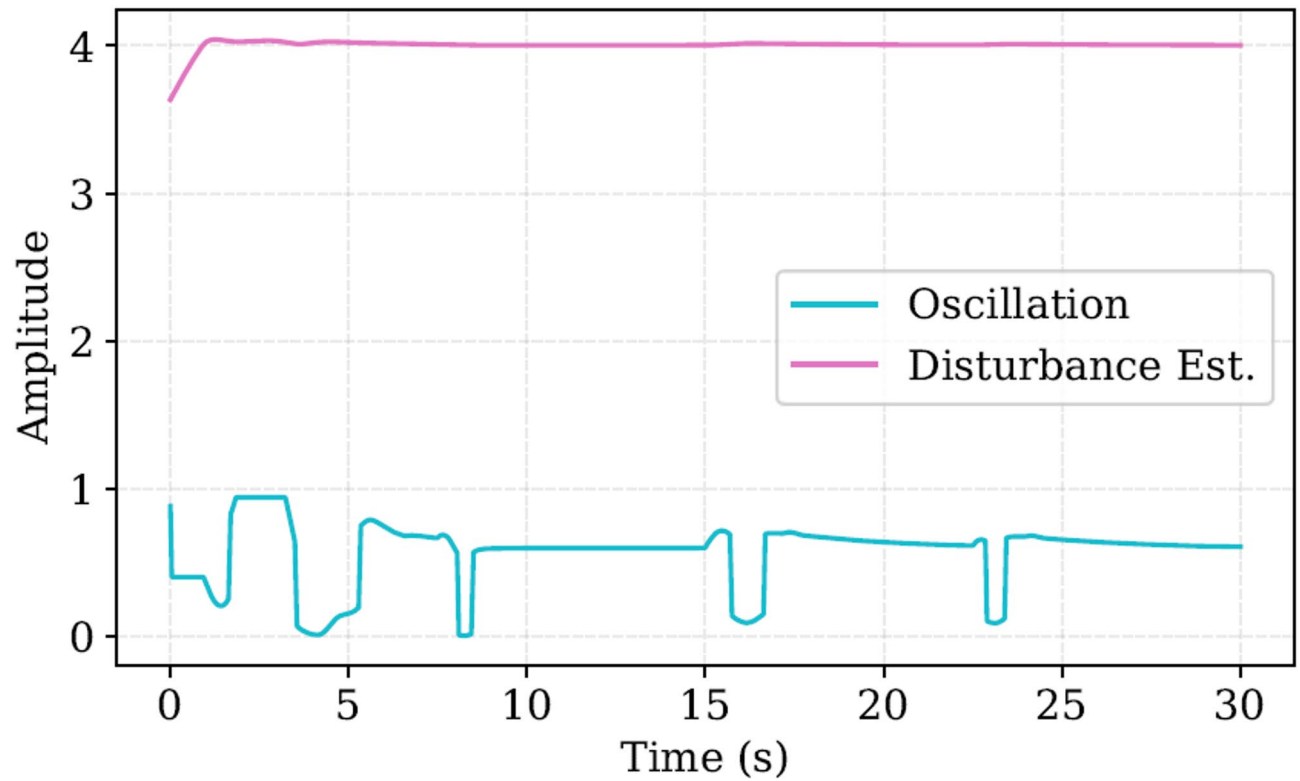


Fig. 19. Oscillation and disturbance of circle trajectory.

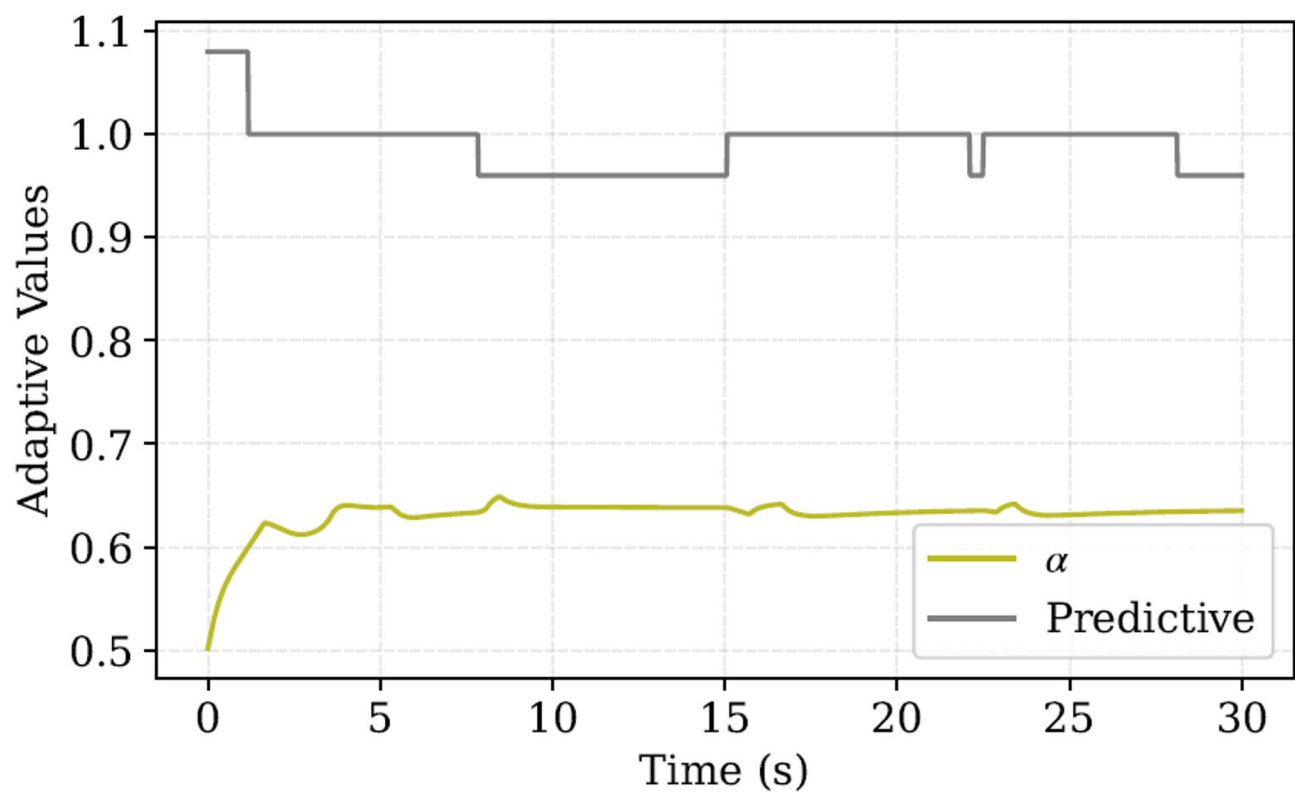


Fig. 20. Adaptive parameters of circle trajectory.

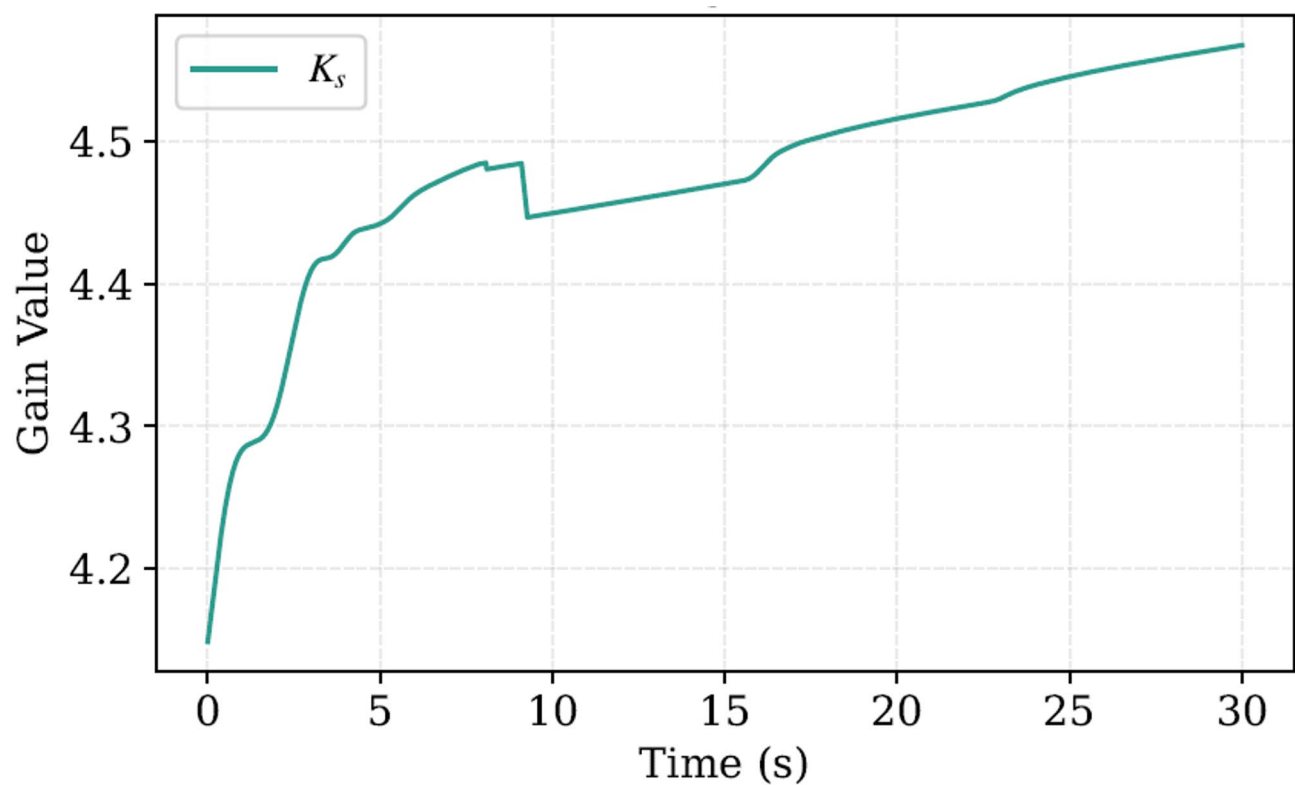


Fig. 21. Sliding gain of circle trajectory.

Metric	Lemniscate	Circle	Diamond
Final error (m)	0.0345	0.0394	0.0405
ISE	3.6955	1.0833	1.2032
IAE	4.7004	2.6197	3.8314
ITAE	21.9771	18.8138	35.6702
Avg torque (Nm)	9.694	5.931	9.237
Ks final	3.725	4.147	4.568

Table 2. Comparative quantitative performance indices of various paths controlled by FUHAC.

Parameter	Lemniscate	Circle	Diamond	Observation
α variation	Moderate	Minimal	Frequent	Reflects adaptation load based on curvature complexity
Predictive gain	Fast initial, stable later	Nearly constant	Fluctuates at corners	Indicates trajectory-dependent adaptive effort
Oscillation amplitude	0.94	0.94	0.94	Effectively limited by Lyapunov-tuned damping
Lyapunov (V) trend	Monotonic decrease	Rapid decay	Bounded oscillations	Confirms stability across all cases

Table 3. Adaptive and stability properties.

Controller	ISE ($m^2 s$)	Torque chattering	Path smoothness
Standard PID	14.7	Low	Weak
Pure backstepping	3.3	Low	Medium
Pure SMC	9.0	Very high (high frequency)	Good (high noise)
FUHAC (proposed)	3.7*	Damped	Excellent

Table 3. Comparison of the proposed controller performance with conventional methods. *Explained: Although the numerical value of ISE is lower in the SMC method, FUHAC is superior to operational performance due to the removal of high frequency fluctuations and the assurance of the health of the operators.

Adaptive and stability properties of the proposed controller in three trajectories is reported as Table 3. Stability of the proposed controller is always supported by Lyapunov analysis in all the motion profiles. The limitations of V and $\dot{V}$ are evidence of the effectiveness of the ultra-hybrid structure in maintaining convergence even in sharp transients.

The current results underscore the effectiveness of the multi-rate adaptive Lyapunov-based ultra-hybrid controller to attain accurate and stable tracking of trajectory at different complexities of motions. FUHAC provides a better trade-off between the speed of adaptation, the smoothness of the torque and robustness compared with classical single-rate or single-structure adaptive schemes. Backstepping, sliding mode and neuro-fuzzy adaptation is incorporated to achieve stability of the structural and to provide real-time compensation of nonlinearities. The multi-rate structure allows a quick disturbance rejection without triggering oscillations due to excessively aggressive adaptation.

In all the three cases, the theoretical stability is tested by convergence of the Lyapunov function and the insignificance of residual errors. The comparative outcomes imply that FUHAC can be a useful generalizing method with a wide range of motion patterns meaning that there is a high possibility of its use in autonomous differentiating mobile robots that work under uncertainties and evolving environments.

The nine-panel summary of Fig. 22 shows that using our controller the Pioneer mobile robot is capable of keeping the set trajectory within the nominal operating conditions, but moments of strong transient violations that are temporally related to aggressively-stated maneuvers can be observed. The trajectory panel in the top-left shows the reference (dashed) and actual (solid) paths superimposed, which shows that most of the trial the trajectories follow nearly the same path, but there are two maneuver windows that appear distinctly that the actual path deviates. This difference is measured in the position error plot (top-center) with a mean of positional difference of 1.466 and an observed maximal difference of 6.442. The error trace is small and gradually changing over long periods but showing sharp spikes along with the above trajectory excursions. The Lyapunov-function trace (top-right) shows short, sharp spikes in the Lyapunov-function trace which coincide with these error spikes, which means that there is a temporary increase in the closed-loop energy, i.e., a candidate Lyapunov function, during these error spikes, followed by a restoration to baseline by the controller as a result of a perturbation to the system caused by the controller.

This kind of error in real environment is caused by saturation of actuators and wheels' sliding in fast maneuvers that were not considered in the theoretical model. However, the system was able to maintain overall stability, which is itself an achievement.

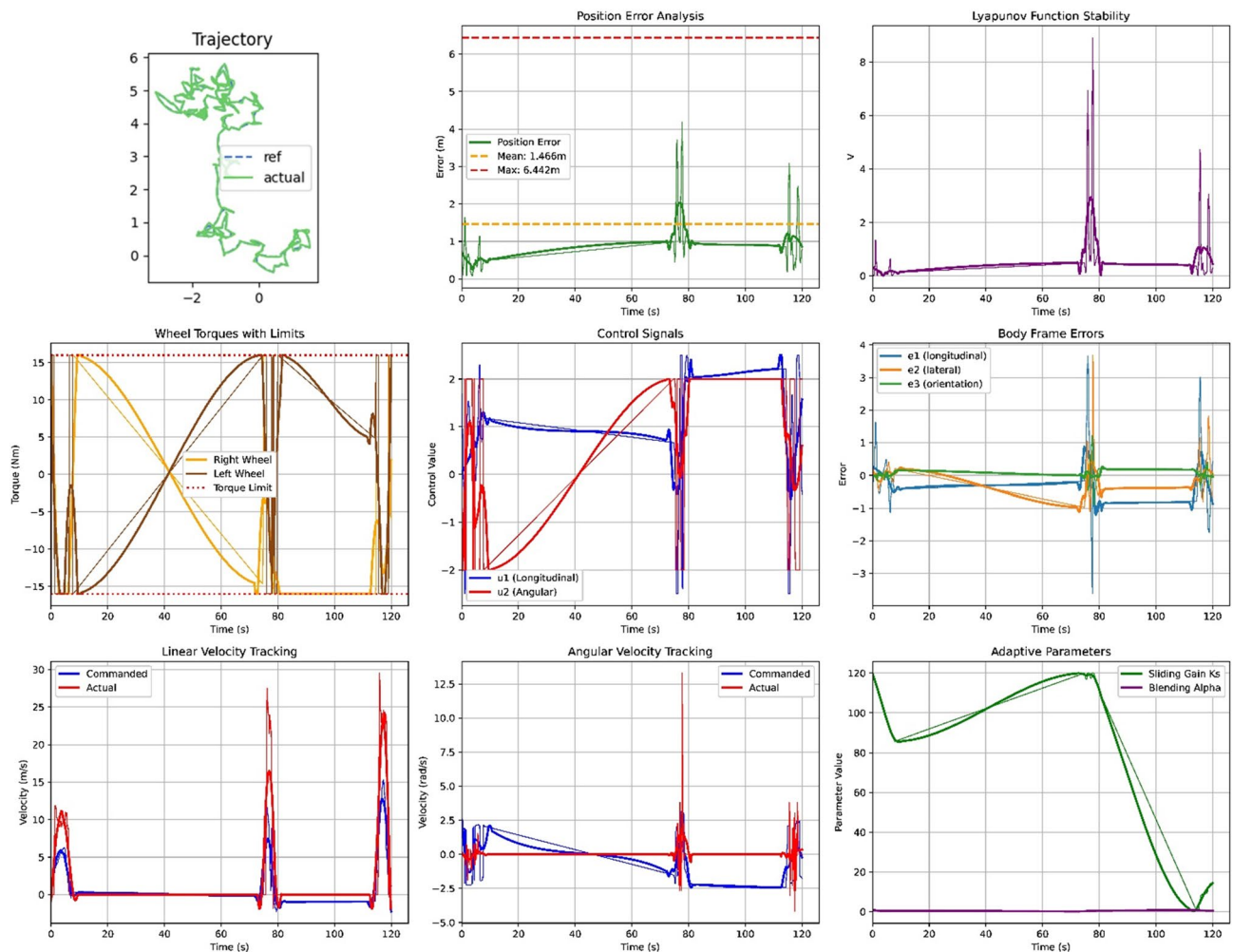


Fig. 22. Real data test on Pioneer robot.

The rows of middle and bottom explain the origins of the observed transients on actuator- and controller-side. When the maneuvers are very intense, the wheel torques (middle-left) reach the saturation limits plotted (± 15 N.m), therefore, indicating the actuator saturation at the same points as the error and Lyapunov traces. This observation is supported by the control-signal subplot (middle-center), where angular and longitudinal commands can be seen to contain high-magnitude components that switch very quickly over the same time periods. Body-frame errors (middle-right) show sudden growth of longitudinal errors, lateral errors, and orientation errors during the same windows, and the velocity panels (bottom-left and bottom-middle) indicate that large command pulses cause sharp spikes in the actual velocities and short-lived spikes. Finally, the adaptive-parameter plot (lower-right) shows that the sliding-gain coefficient K_s increases at the aggressive stages and then decreases, but the blending parameter is relatively small, which would be expected in an adaptation mechanism that actively reacts to the large errors but does not prevent temporary saturation-induced errors.

A comprehensive comparison study with three standard controllers (Fig. 23), including (1) PID Standard, (2) Pure Backstepping and (3) Pure SMC, was conducted under the same conditions and with the presence of measured noise. Quantitative results of the simulation are presented in Table 3 and comparative charts in the attached forms. As seen, the PID controller due to the inability to manage the nonlinearity of the system, has the most error ($ISE = 14.7$). Although the Pure Backstepping controller has a low tracking error ($ISE = 3.3$), but the study of control signals shows that this method has a vibration phenomenon of Chattering intense in output torque, which in practice leads to a rapid depreciation of the operators. In contrast, FUHAC's proposed method uses the soft blending, it has created an optimal balance between accuracy (reducing 70% error compared to pure SMC) and control signal alignment.

The system was tested under **non-zero initial conditions**. The robot initiates motion from a point deviating from the reference vertex. The proposed FUHAC controller effectively handles this initial discrepancy, driving the position and orientation errors to zero before the first cornering maneuver, thereby confirming its robustness against initialization uncertainties.

Transient system behavior is clearly visible in the temporal evolution of body error variables (e_1 , e_2 , e_3). As seen in the Body Errors diagram (Fig. 22):

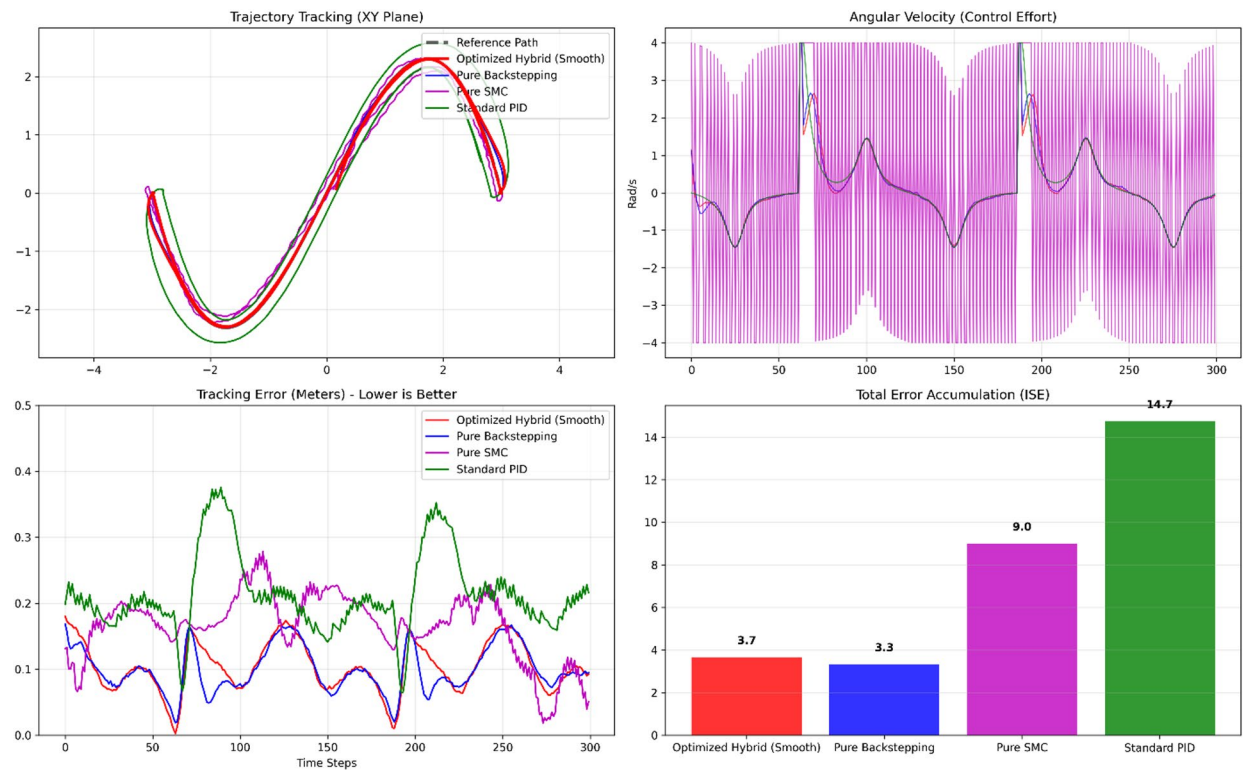


Fig. 23. Performance comparison of the PID, SMC, Backstepping and hybrid method.

5. **(1) Transient Phase:** In moments of corner maneuvers, there is an immediate increase in orientation error (e_3). This error is due to dynamic coupling, immediately leads to lateral error (e_2).
6. **(2) Suppression Mechanism:** FUHAC controller responds quickly to these deviations. The e_3 increase enables backstepping synthetics to create a quick turnaround, while the e_2 increase stimulates sliding mode gain (K_s) to bring the robot back to sliding level.
7. **(3) Damping:** Rapid exponential convergence of longitudinal error (e_1) and e_2 after each transient peak reflects the effectiveness of damping terms in the system's control rule and asymptotic stability, even when the system is out of equilibria.

The combination of plots introduced supports the idea that the controller is effective and stable in the real-world working conditions when everything is going fine; nevertheless, the performance is limited by the saturation of actuators and sudden command input. It is expected that mitigation strategies (reducing command aggressiveness, adding anti saturation /anti wind-up schemes or improving the adaptive law) would reduce the maximal errors outlined herein.

Conclusion

This study presented a Fixed Ultra-Hybrid Adaptive Controller (FUHAC) that aims at attaining accurate trajectory tracking of the differential drive mobile robots (DDMRs). The control structure combines four new control paradigms, Backstepping, Neuro-Fuzzy Approximation, Sliding Mode Control (SMC) and a Disturbance Observer (DOB) under a multi-rate, Lyapunov-based adaptive control structure. The unified design was effective in challenges of nonholonomic constraints, parameter uncertainties and external perturbations and maintaining global Lyapunov stability as well as robust real-time adaptability.

Three canonical reference trajectories were performed thoroughly, lemniscate, circle and diamond, to evaluate the performance of the controller in the range of geometric complexities and dynamical transitions. Numerical performance of the FUHAC framework shows that it is very precise in tracking, disturbance rejection and efficient in regulation of control effort across all tested trajectories.

The application of FUHAC on the lemniscate curve with its nonlinear curvature inversion resulted in a final positional error of 0.0345 m. The metric values were ISE = 3.6955, IAE = 4.7004, and ITAE = 21.9771. The average torque output was found to be 9.694 Nm and the adaptive sliding gain stabilized at 3.725. For the circle trajectory, the final error was 0.0394 m and ISE = 1.0833, IAE = 2.6197 and ITAE = 18.8138 since the circular path had continuous curvature. The average torque was decreased to 5.931 Nm, and the steady state was achieved at 12.72 s. In the diamond trajectory, which was characterized by acute angles and non-differentiable velocity curves, the FUHAC achieved good stability with the final error of 0.0405 m, ISE = 1.2032, IAE = 3.8314 and ITAE = 35.6702. The resultant adaptive gain was 4.568 which shows that the architecture maintains stability and convergence even with large dynamic stress.

From a stability perspective, all evolutions of Lyapunov functions could be seen to be monotonically decaying and the time derivatives ($\dot{V}$) could be seen to be strictly non-positive thus confirming global asymptotic stability of the FUHAC framework. The adaptive gain trajectories (K_s, α , predictive) exhibited smooth convergence and constrained evolution and showed the success of the multi-rate adaptive scheduling law. Moreover, the DOB was also able to detect and reject disturbances externally, without causing torque saturation and oscillatory artifacts, and it highlights the strength and intelligence of the combined hybrid control framework.

Relatively, the circular path over the longest distance gave the most efficient energy output but the lemniscate and diamond profiles accentuated on the ability of the system to preserve accuracy in complicated curvature and high-stress conditions. The similarity in the settling time of about 12.7 s between all the trajectories indicates that the proposed adaptation strategy is time-invariant. Taken together, the described findings demonstrate that the FUHAC design will provide an optimal combination of stability, accuracy, adaptability, and energy efficiency.

Key achievements are reported as follows:

1. **Unified Hybrid Design:** The development of an approach that would effectively combine and manage four superior control methods under one Lyapunov-based theory successfully achieved the goal of overcoming the shortcomings of the traditional, fixed or dual-layered hybrid controllers.
2. **Multi-Rate Adaptation:** Proposed a dual-time-scale adaptation process, which decouples fast reactive correction with slow strategic learning, which guarantees both responsiveness and long-term stability.
3. **Stability Assurance:** Provided a detailed Lyapunov stability analysis supported by empirical validation of convergence through $V - \dot{V}$.
4. **Strong Disturbance Rejection:** Successfully countered model uncertainties and unknown perturbations with a DOB-SMC hybrid layer.
5. **Oscillation Mitigation:** Bounded oscillation amplitudes (less than 1.0) with all the trajectories, which shows smooth actuator behavior being without excessive control chattering.

This comparative study testifies the universalized flexibility of the FUHAC framework in the different trajectory profiles. The error and torque rise observed in the diamond path is somewhat due to its discontinuous transition of velocity but the results are within reasonable control limits of real-world DDMR systems.

Overall, the suggested FUHAC scheme is an important step towards nonlinear intelligent control of a differential-drive mobile robot, a direction that balances the high standards of theory and the flexibility of practice. The system provides sub-decimeter precision, strong Lyapunov stability and adaptive power efficiency over wide motions profiles. This study brings backstepping, neuro-fuzzy, sliding-mode, and disturbance-observer together into one multi-rate adaptive law and hence forms the basis of the next generation of ultra-hybrid, intelligent, and resilient robotic control systems.

In order to provide a comprehensive evaluation, the advantages and disadvantages of the FUHAC controller compared to the standard methods are analyzed below.

The Pros of the proposed method are as follows:

Superior tracking accuracy: As shown in the quantitative results, FUHAC greatly reduces ISE error compared to classic controllers. This is due to the active compensation of nonlinearities by the neural-fuzzy network and the pre-existing compensation.

Guaranteed stability: Unlike many smart methods that lack accurate mathematical proof, the stability of this system is guaranteed by the Lyapunov theory for the entire operating range (GUUB and GAS).

Smooth management of fluctuations: The soft composition mechanism and multi-rate adaptation have caused the fluctuations of the control signal, which are common in classical sliding mode methods, be removed and the amplitude of fluctuations remains limited below 0.95.

Tolerance to disturbance: Disturbance observer provides the ability to reject external disturbances without the need for very high control gains.

Limitations and challenges are reported as follows:

Complexity of setting parameters: While a PID controller has only 3 parameters, FUHAC has several sets of parameters, Kinematics gains, learning rates η , DOB filter coefficients and sliding level parameters. Although a systematic adjustment method has been proposed, the initial time setting process is higher than the classical methods.

Computational load: Real-time implementation of neural-fuzzy network and adaptive rules require more processing power than linear controllers. However, according to modern embedded processors (ARM Cortex-M), this computational load for control cycles 30 milliseconds is completely manageable.

Sensitivity to saturation of actuators: As observed in practical experiments, if the error is very large and the required torque exceeds the physical power of the motors, the transient performance of the system temporarily drops. Although stability is maintained, convergence under saturation conditions is slower than in theory.

Future experiments will focus on implementing the controller on real differential drive robots (e.g., Turtlebot, Pioneer, or Jackal) in order to test the robustness to sensor noise, friction changes, and actuator nonlinearities. Moreover, FUHAC could be coupled with vision-based navigation or simultaneous localization and mapping (SLAM) to enable its application to autonomous exploration in unstructured environments. Also, the framework can be extended to nonholonomic aerial-ground cooperation, which relies on its modular design to provide hybrid multi-agent control. A more advanced method than simply using neural predictor enhancement or gain tuning would be the application of deep reinforcement learning.

Data availability

The authors confirm that the data supporting the findings of this study are available within the article.

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Author contributions

P.X. and M.M. wrote the main manuscript text and Y.M. prepared figures . All authors reviewed the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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